

Probabilistic “Copies” in Generative AI Models

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Recent work shows that it is possible to extract verbatim or near-verbatim text of some copyrighted works from some large language models (LLMs or models). That is evidence that the model weights encode the works in some form – that the model has “memorized” those works from its training data. But LLMs don’t store information in the same format as familiar databases. Rather, their weights store statistical relationships between tokens that have been learned from the training data, and those relationships inform a generation process that is often probabilistic rather than deterministic. In the case of memorization, those relationships are strong enough that, in many circumstances, the model might generate a copyrighted work from its training data with some probability

Copyright law has not previously had to decide whether storing information that might or might not produce output similar to a copyrighted work is itself a copy of the work. The answer to the question is important, because it may determine the legality of many LLMs. The statute and case law are largely unhelpful. We argue that copyright law will likely take a functional approach to the question, finding that LLMs contain a copy of a particular work only if it is straightforward to extract that work in outputs. That result is unsatisfying as a policy matter, and we suggest potential changes to the law, but it is the most likely outcome under current law.

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More than one hundred pending lawsuits assert that generative AI infringes copyright law.¹ Most of those suits challenge the legality of training AI on copyrighted works.² An increasing number allege infringement because the output of AIs is the same as or similar to a copyrighted work³ – for instance, a near-exact reproduction of specific song lyrics.⁴

Until November 2025, courts and litigants hadn’t paid much attention to what happens in between training the model and generating outputs – the copyright status of the model itself. A few early U.S. suits claimed the models were derivative works simply because they were trained on copyrighted works. Courts generally rejected those claims.⁵ But two recent decisions – one in

1. AtkinsonAILaw AI Litigation Tracker, <https://docs.google.com/spreadsheets/d/1BgoFqE0C6148M2evzZU9A-WJN0oys7Pm/edit?usp=sharing&ouid=100490625071723772258&rtopof=true&sd=true>; see also ChatGPT is Eating the World, <https://chatgptiseatingtheworld.com/>.

2. See, e.g., *Bartz v. Anthropic PBC*, 787 F. Supp. 3d 1007, 1034 (N.D. Cal. 2025) (holding that copying of works to train LLMs is fair use); *Kadrey v. Meta Platforms, Inc.*, 788 F. Supp. 3d 1026, 1057 (N.D. Cal. 2025) (holding that copying of works to train LLMs is fair use, but emphasizing that insufficient evidence was presented by the plaintiffs on the effect of use upon the potential market for the copyrighted works); *Thomson Reuters Enter. Ctr. GmbH v. Ross Intel. Inc.*, 765 F. Supp. 3d 382, 401 (D. Del. Feb. 11, 2025) (holding that copying of works to train a legal research platform on Reuters’ headnotes was not protected by fair use primarily because Ross was developing a market substitute).

3. See, e.g., *New York Times Co. v. Microsoft Co.*, 777 F. Supp. 3d 283, 307 (S.D.N.Y. 2025) (denying motion to dismiss plaintiffs’ claim that defendants’ LLMs outputs were substantially similar to plaintiffs’ works); *In re OpenAI, Inc. Copyright Infringement Litigation*, No. 25-md-3143, 2025 U.S. Dist. LEXIS 211544, at *108 (S.D.N.Y. 2025) (declining to dismiss plaintiffs’ claims that some model outputs were substantially similar to plaintiffs’ works); *Tremblay v. OpenAI, Inc.*, 716 F. Supp. 3d 772, 778 (N.D. Cal. 2024) (dismissing plaintiffs’ claim that every output of the model trained on their copyrighted materials is an infringing derivative work).

4. *GEMA v. OpenAI*, 42 O 14139/24 (LG Munchen 11 Nov. 2025) (holding that reproduction of song lyrics in model outputs was infringing).

5. See *Andersen v. Stability AI Ltd.*, 700 F. Supp. 3d 853, 865 (N.D. Cal. 2023) (dismissing plaintiffs’ claim that the model itself is an infringing derivative work); *Kadrey v. Meta Platforms, Inc.*, 788 F. Supp. 3d 1026, 1057 (N.D. Cal. 2025) (dismissing plaintiffs’ claim that the model itself is an infringing derivative work).

Germany and one in the U.K. – have brought the question of whether the model itself incorporates a copy of the works it was trained on to the fore. One decision found that ChatGPT incorporated a copy of copyrighted song lyrics, which appeared in outputs.⁶ The other, involving the image-generation model Stable Diffusion, found the opposite.⁷ U.S. courts have not yet confronted the issue of whether the model is a copy.⁸

The question is likely to grow in importance, and touches on a line of technical research that the scientific literature refers to as “memorization.” We address the relevant details of this term in Part I, but the intuition is as follows: during training, specific training data leave a sufficiently strong imprint on the model’s weights such that it can be possible to reproduce — i.e., extract — those training data in the model’s outputs. In other words, “extraction” is an observable consequence of memorization: if a piece of training data is memorized, it may be possible to extract it from the model.⁹

Researchers in machine learning have studied both memorization¹⁰ and extraction¹¹ for years, long before generative AI attracted attention from

6. See *GEMA v. OpenAI*, 42 O 14139/24 (LG Munchen 11 Nov. 2025); Jorn Poltz & Friederike Heine, *OpenAI Used Song Lyrics in Violation of Copyright Laws, German Court Says*, Reuters (Nov. 11, 2025), <https://www.reuters.com/world/german-court-sides-with-plaintiff-copyright-case-against-openai-2025-11-11/> (summarizing some of the plaintiff material from *GEMA v. OpenAI*).

7. See *Getty Images (US) Inc & Ors v. Stability AI Limited* [2025] EWHC 2863 (Ch) (holding that Stability AI’s model was not itself an infringing copy of Getty’s images).

8. Some courts have rejected claims that the model is by definition a derivative work of anything it is trained on. See *Andersen*, 700 F. Supp. at 861.

There is a natural relationship between the training of a model and any content that is memorized in the model. But the training cases do not present the question of whether a copy has been made in the model; rather, they concern the intermediate copy which defendants concede is contained in the training data, and defend as fair use.

9. See A. Feder Cooper & James Grimmelman, *The Files Are In the Computer: On Copyright, Memorization, and Generative AI*, 100 Chi.-Kent L. Rev. 141, 142 (2025); A. Feder Cooper et al., Report of the 1st Workshop on Generative AI and Law 30 (Dec. 3, 2023) (unpublished manuscript) <https://arxiv.org/abs/2311.06477> [hereinafter “GenLaw Report”]. Memorization is a technical term in the scientific literature on machine learning, not one that we have picked to describe the phenomenon in question. In many respects, the choice of terminology is an unfortunate one, as it sounds like a metaphor for how AI models behave – that they “memorize” information in a way like a human might when studying for an exam. But this term is not a metaphor, and it isn’t meant to invoke this meaning: it is shorthand for the specific technical phenomenon that we provide an intuition for here, and for which we provide more background details below. See *infra* Part I.

10. See, e.g., Vitaly Feldman, *Does Learning Require Memorization? A Short Tale about a Long Tail* (Jan. 10 2021) (unpublished manuscript), <https://arxiv.org/abs/1906.05271>; Reza Shokri et al., *Membership Inference Attacks against Machine Learning Models* (Mar. 31, 2017) (unpublished manuscript), <https://arxiv.org/abs/1610.05820>.

11. See, e.g., Nicholas Carlini et al., *Extracting Training Data from Large Language Models*, in SEC ‘21: Proceedings of the 39th USENIX Conference on Security Symposium 5253 (USENIX Association, 2023) <https://www.usenix.org/system/files/sec21-carlini-extracting.pdf> [hereinafter “Extracting training data from LLMs”]; Katherine Lee et al., *Deduplicating Training Data Makes Language Models Better*, in Proceedings of the 60th Annual Meeting of the Association for

copyright law. But the training of commercial AI systems on copyrighted works has made memorization a copyright question, not just a machine learning one. Academic research by ourselves and others has shown that it's possible to extract at least some copyrighted works verbatim from AI models.¹² More recent work shows that, even if a copyrighted work can't be extracted verbatim from an AI model, the model can sometimes reproduce near-verbatim copies in its outputs.¹³ These copies are sufficiently similar that they would infringe copyright, which does not require identical duplication.¹⁴

These findings have implications for AI outputs, but also potentially for the models themselves. As is well-established in the AI scientific community, extraction is evidence that the memorized training data is encoded in some form in the model's weights.¹⁵ Writing about this for a legal audience, Cooper and

Computational Linguistics (Volume 1: Long Papers) (Association for Computational Linguistics, 2022) <https://aclanthology.org/2022.acl-long.577/> [hereinafter "Deduplicating"].

12. See Milad Nasr et al., Scalable Extraction of Training Data from (Production) Language Models 9 (Nov. 28, 2023) (unpublished manuscript), <https://arxiv.org/abs/2311.17035>; A. Feder Cooper et al., Extracting Memorized Pieces of (Copyrighted) Books from Open-weight Language Models 14 (May 1, 2026) (unpublished manuscript) <https://arxiv.org/abs/2505.12546> [hereinafter "Extracting Books (Open-Weight)"]; Antonia Karamolegkou et al., *Copyright Violations and Large Language Models*, in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing 7403, 7406 (Association for Computational Linguistics, 2023) <https://aclanthology.org/2023.emnlp-main.458/>.

13. See Ahmed Ahmed et al., *Extracting Books from Production Language Models* XX (Jan. 6, 2026) (unpublished manuscript), <https://arxiv.org/abs/2601.02671> [hereinafter "Extracting Books (Production LLMs)"]; Guangwei Zhang et al., *Copyright Detective: A Forensic System to Evidence LLMs Flickering Copyright Leakage Risks* (Feb. 11, 2026) (unpublished manuscript), <https://arxiv.org/abs/2602.05252>; A. Feder Cooper et al., *Estimating Near-Verbatim Extraction Risk in Language Models with Decoding-Constrained Beam Search* XX (Mar. 26, 2026) (unpublished manuscript), <https://arxiv.org/abs/2603.24917> [hereinafter "Near-Verbatim"].

14. "An infringement is not confined to literal and exact repetition or reproduction; it includes also the various modes in which the matter of any work may be adopted, imitated, transferred, or reproduced, with more or less colorable alterations to disguise the piracy." *Universal Pictures Co. v. Harold Lloyd Corp.*, 162 F.2d 354, 361 (9th Cir. 1947). The test for copyright infringement is substantial similarity, though how that similarity is tested can differ by circuit. The Ninth Circuit, for example, uses a two part test: an extrinsic test based on objective similarity, and an intrinsic test based on the response of a lay observer. See *Sid & Marty Krofft Television Prods., Inc. v. McDonald's Corp.*, 562 F.2d 1157, 1164 (9th Cir. 1977). The Second Circuit, similarly, asks "whether defendant took from plaintiff's works so much of what is pleasing to the ears of lay listeners." *Arnstein v. Porter*, 154 F.2d 464, 473 (2d Cir. 1946). See Mark A. Lemley, *Our Bizarre System for Proving Copyright Infringement*, 57 J. Copyright Soc'y 719 (2010) (discussing the various tests).

15. See generally Extracting training data from LLMs, *supra* note 13; Nicholas Carlini et al., *Extracting Training Data from Diffusion Models*, in SEC '23: Proceedings of the 32nd USENIX Conference on Security Symposium 5253 (USENIX Association, 2023), <https://dl.acm.org/doi/10.5555/3620237.3620531> [hereinafter "Extracting Training Data from Diffusion Models"]; <https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting>; Nicholas Carlini, *What My Privacy Papers (Don't) Have to Say About Copyright and Generative AI* (Mar. 11, 2025), <https://nicholas.carlini.com/writing/2025/privacy-copyright-and-generative-models.html> [hereinafter "Carlini Blog"]; Avi Schwarzschild et al., *Rethinking LLM Memorization Through the Lens of Adversarial Compression*, in Proceedings of the 38th International Conference on Neural Information Processing Systems (Curran Associates Inc., 2024), <https://arxiv.org/abs/2404.15146>; Nasr et al., *supra* note 12 at XX; Milad Nasr, *Scalable Extraction of*

Grimmelmann note that the fact that a model can generate long verbatim snippets of a book (e.g., text from *Harry Potter*) means that at least those portions of the book are encoded in the model itself.¹⁶ And the Copyright Office has declared, without case support, that because model weights are fixed in a tangible medium, anything contained in them is also a fixed copy.¹⁷

But the *extent* and the *form* in which that encoding happens are likely to matter a great deal under existing copyright law. If an AI model contains a “copy” of a copyrighted work as copyright law defines that term, and it merely replicates that copy in its outputs on request, the creation of the model itself is an act of direct infringement; so is the making of a new copy of that model – for example, downloading the weights of an openly released model, like one from Meta’s Llama 3 series, to a corporate server. That is true even if no one ever asks for – and the model never generates – infringing output.¹⁸ By contrast, if the model contains only information that could be used to produce a copy and generates the output only in response to particular prompts – assembling the output on the fly from sources that do not include a legally cognizable copy of the work itself – the output may be an infringing copy but the model itself likely wouldn’t be.

Determining which of these descriptions fits AI models turns out to be surprisingly complicated. The difficulty lies both in the technical details of how models encode information and generate outputs and in copyright law’s existing treatment of copies and fixation. And so the rest of this Article proceeds in two steps. Part I provides a minimal account of how language models encode information, how “memorization” reflects a particular degree of encoding of training data in the model’s weights, and why ordinary metaphors like storage, memory, and compression both illuminate and mislead. The point is not to settle the legal questions by analogy, but to show why the relevant copyright questions cannot be resolved at the level of abstraction those analogies invite. Part II then turns to copyright law, examining whether models that can (deterministically or probabilistically) generate copyrighted works from their training data should be treated as fixed copies, and what follows if they are.

Training Data from (Production) Language Models XX (Nov. 28, 2023) (unpublished manuscript), <https://arxiv.org/abs/2311.17035>, Giorgio Franceschelli, Claudia Cevenini & Mirco Musolesi, *Training Foundation Models as Data Compression: On Information, Model Weights and Copyright Law* XX (Mar. 12, 2025) (unpublished manuscript), <https://arxiv.org/abs/2407.13493>.

16. Cooper & Grimmelmann, *supra* note 9, at 168.

17. United States Copyright Office, *Copyright and Artificial Intelligence Part 3: Generative AI Training* 29 (2025). See also Daniel Gervais et al., *The Heart of the Matter: Copyright, AI Training, and LLMs*, 71 J. Copyright Soc’y 485, 494–98 (2025); Joshua Yuvaraj, *Pseudo-Expressions: An Expanded conception of the Work in Copyright Law* (working paper 2025) (both arguing that model weights can be copies).

18. The copy in the model in this scenario may still qualify as fair use; more on that below.

I. THE TECHNICAL COMPLICATIONS OF MEMORIZATION AND EXTRACTION

What is often called “generative AI” refers to a set of technological systems that resist simple description. It’s therefore natural to reach for metaphors and analogies — to use the familiar to try to make sense of things that are complex and unfamiliar. This is an especially common pattern in law: “Analogy and metaphor are central to legal rhetoric; they provide a rational framework for thinking through the relevant similarities and differences between cases.”¹⁹ But “metaphors . . . can also simplify and distort,”²⁰ and so legal practitioners (as well as others) “should beware of metaphors that provide too-easy answers to the genuinely hard problems before them.”²¹

This warning is critical to keep front of mind when thinking about what the machine learning literature refers to as “memorization,” as well as related terms such as “compression” and “training.” These terms all invoke familiar, everyday meanings, but in machine learning they also refer to precise technical concepts.²² A central challenge in work at the intersection of generative AI and law is that the scientific literature is rife with terms that do this metaphorical-technical double duty: shorthand labels that gesture toward, but also obscure, the underlying technical mechanisms they are meant to describe.²³ If one, ironically, leans too literally into the metaphorical side, it can be tempting to reject these terms as “fallacies” or failures of abstraction.²⁴ But we can’t understand the accuracy of the metaphor without the technical substance those terms are actually referring to.

At the risk of making our own imperfect analogy, law faces similar terminological challenges. Many core legal concepts are easily misunderstood by non-experts because they carry ordinary meanings that differ from their technical legal definitions. In copyright, “fair use,” “transformative,” and even “copy” are three such concepts that invite confusion when invoked on non-technical terms. But the solution in law is not to throw out these terms – to call

19. GenLaw Report, *supra* note 9, at 4. *See also* Daniel Solove, *Privacy and Power: Computer Databases and Metaphors for Information Privacy*, 53 Stan. L. Rev. 1393, 1397–99 (2001) (arguing that the database problem cannot be adequately understood through the use of the big brother metaphor).

20. GenLaw Report, *supra* note 9, at 4. *See also* Appendix B; Katherine Lee, A. Feder Cooper & James Grimmelmann, *Talkin’ ‘Bout AI Generation: Copyright and the Generative-AI Supply Chain*, 72 J. Copyright Soc’y 251 Part III (2025) [hereinafter “Talkin’”]. For a longer and more recent treatment of related issues, *see generally* Michiel A. Smit, *Metaphors we judge (AI) by: a rhetorical analysis of artificial copyright disputes*, 21 Journal of Intellectual Property Law & Practice 250 (2026).

21. *See* introduction of Talkin’, *supra* note 20.

22. GenLaw Report, *supra* note 9, at 4–5.

23. *See* Michiel A. Smit, *Metaphors we judge (AI) by: a rhetorical analysis of artificial copyright disputes*, J. Intell. Prop. L. & Prac. (2026).

24. *See infra* Part I.C.

them “fallacies” because they are frequently misunderstood by others – but to engage them thoughtfully on their technical merits to make sound, reasoned arguments.

So in that spirit, the remainder of this section develops a minimal technical account of language models and “memorization” – the imprint left in the model’s weights by a specific work in the training data, sufficient to encode part or all of a copyrighted work in the model itself. We focus on language models because they have been the subject of our own empirical work in machine learning, from which we draw clear examples of memorization.²⁵ ²⁶ We don’t attempt to provide a comprehensive treatment of memorization, and instead defer to others.²⁷ Rather, we aim to clarify the specific mechanisms necessary to understand the relationship between memorization – which is a property of a trained model itself – and when a model produces copies of its training data as outputs. Memorization is a complex phenomenon, for which the scientific literature continues to evolve. One of the key takeaways from the literature is that it isn’t possible to make broad, one-size-fits-all statements about memorization and AI. We therefore also

highlight where popular metaphorical understandings of memorization fall short, and show why the relevant questions for copyright can’t be resolved at the level of abstraction that metaphors and analogies encourage. This provides the foundation for our copyright analysis in Part II.

A. How Language Models Store and Decode Information

Language models are made up of a large collection of numerical parameters, commonly referred to as weights. These weights are adjusted during training,²⁸ a process that takes as inputs both a model architecture (how those weights are structured and interact) and a training dataset. The result is a trained

25. While we focus on text and LLMs, memorization also arises in other forms of generative AI, such as images and software code. See *Extracting Training Data*, *supra* note 15, at XX (extracting images from Stable Diffusion); Aneesh Pappu et al., *Measuring Memorization in RLHF for Code Completion*, in *The Thirteenth International Conference on Learning Representations* (2025), <https://openreview.net/forum?id=Tg8RLxpMDu> (studying memorization in code completion). We particularly focus on *autoregressive* LLMs; while other LLMs, such as diffusion language models (DLMs) are growing in popularity, autoregressive LLMs currently remain the dominant paradigm.

27. See generally Lee et al., *supra* note __, at Part II.C (discussing memorization in detail); Cooper & Grimmelmann, *supra* note 9 (defining memorization for a legal audience).

28. The term “training” also performs the same metaphorical-technical double duty we’ve been talking about. In machine learning, it refers to an optimization process in which a model’s weights are adjusted to minimize a loss function over a dataset, rather than to anything like human learning or instruction. Unlike “memorization” or “compression,” however, “training” is largely understood in terms of its technical meaning in the legal literature. See *GenLaw Report*, *supra* note 9, at XX.

model: a fixed set of weights that encode information derived from the training data.²⁹

This is how models store information³⁰ from their training data, and it's important to be clear about what that means. "Store" here does *not* literally mean that the model retains discrete copies of its training data in exactly the same format as a traditional database or lookup table would.³¹ The information isn't localized or laid out in that way. But it does mean that certain information about the training data is encoded in the model's weights. That encoded information determines the probability distribution the model assigns to sequences of tokens that represent text. For example, after seeing the phrase "The dog barked at the" the model may assign a higher probability to the next token being "cat" compared to the next token being "chair" or "moon," reflecting what it has learned from the training data about the relationships between words.

Beyond this high-level description, we in fact know surprisingly little about how AI models actually store information. For a text file stored in a computer, one could point to a particular set of 1s and 0s and say "that's the data that corresponds to this particular file." That's generally not true for AI models. To be clear, the problem is *not* that the AI companies know this information and have concealed it from the world. It's that there remain big, open scientific questions on this topic: even the people who build and train models typically don't know precisely what pieces of information models retain or where exactly they store it.³²

29. See Talkin', *supra* note 20, Part I.A and I.B for a treatment of background information on data and machine learning addressed to a legal audience.

30. Like "memorization" and "compression," the claim that language models "store" information is not merely metaphorical. In a technical sense, a trained model encodes information about its training data in its weights, which parameterize a probability distribution over token sequences. This concept can be formalized using tools from information theory: predictive models induce compression schemes in which better prediction corresponds to more efficient representation of the underlying data in the model's weights. See, e.g., Grégoire Delétang et al., *Language Modeling Is Compression* in The Twelfth International Conference on Learning Representations (2024), <https://openreview.net/forum?id=jznbginyus>. Of course, this notion of storage differs fundamentally from discrete, addressable storage in traditional databases or lookup tables, but it's nonetheless a precise and measurable property of the model and so it's correct to refer to this process as "storage." For more on why machine learning models are not at all like traditional databases, see generally A. Feder Cooper et. al., *Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research* (NeurIPS 2025) <https://openreview.net/forum?id=mfd6GRW4Az> [hereinafter "Machine Unlearning"]. For a brief treatment of the relationship between compression and memorization, see generally Franceschelli, Cevenini & Musolesi, *supra* note 15, at XX.

31. The *format* is different from a traditional database or lookup table, but the *behavior* can be rigorously related to a stochastic lookup table. See A. Feder Cooper et al. "Untitled Manuscript" XX (2026) (Unpublished Manuscript) (manuscript on file with the authors) [hereinafter "Cooper Untitled"].

32. Developing a better understanding of these phenomena is an active area of ongoing research in machine learning. See Machine Unlearning, *supra* note 3028, at XX (and citations therein concerning mechanistic interpretability).

Due to this complexity, it's common to describe this whole process as learning “statistical correlations,”³³ “relationships,”³⁴ or “patterns”³⁵ in the training data. That description isn't wrong, but it's also only a very high-level gloss that oversimplifies the information represented by the model. For that matter, the same is true of saying that a model “learns” from its training data. That terminology is useful, but it shouldn't be taken too literally: a language model isn't a human brain, and its internal representation doesn't map exactly onto familiar concepts like human memory or understanding.

This is also the sense in which the term compression is sometimes used in the technical literature. Training a model can be understood as constructing a compact (i.e., compressed) representation of the data that supports accurate prediction.³⁶ This is almost surely what Emad Mostaque, a prior CEO of the Stability AI, had in mind when he said the following in 2022 about the Stable Diffusion image generation model:

It's worth taking a step back and thinking about how crazy insane this is: we took a hundred terabytes of data—a hundred thousand thousand megabytes of images—2 billion of them—and we squished it down to a 2–4 gigabyte file. And that file can create everything that you've seen. That's insane, right? That's about as compressed as you can get.³⁷

Compression is a technically accurate way to describe the results of training to a computer scientist; training a language model is compression in an information theory sense,³⁸ even though a model is not arranged or otherwise formatted like more familiar data formats, such as MP3 or PDF files.³⁹ But what information theory means by compression may not be what lay users have in mind when they hear the term.⁴⁰

Once trained, a language model can be used to generate text. Given an input prompt, an autoregressive LLM predicts the next token in a sequence based on the tokens that precede it, then repeats this process iteratively to produce longer

33. Def. Anthropic PBC's Opposition to Plaintiffs' Motion for Preliminary Injunction at 4, *Concord Music Grp., Inc. v. Anthropic PBC*, No. 3:23-cv-01092 (M.D. Tenn. Jan. 16, 2024).

34. *Bartz v. Anthropic*, 787 F. Supp. 3d 1007, 1017 (N.D. Cal. 2025).

35. *Concord Music Grp. v. Anthropic PBC*, 772 F. Supp. 3d 1131, 1134 (N.D. Cal. 2025).

36. See Delétang et al., *supra* note __, at 1–3. More on this below, see *infra* Part I.C.

37. See First Amended Complaint at 27, *Andersen v. Stability AI, Ltd.*, No. 3:23-cv-00201 (N.D. Cal. Nov. 29, 2023).

38. See Delétang et al., *supra* note __ at XX; Franceschelli, Cevenini & Musolesi, *supra* note 15, at XX.

39. See Cooper & Grimmelmann, *supra* note 9, at XX.

40. Matthew Sag, *Copyright's Jagged Frontier*, X Duke L.J., (forthcoming Feb. 28, 2026) (manuscript at 36), <https://ssrn.com/abstract=6319379>.

outputs.⁴¹ Each prediction is probabilistic: based on the “patterns” the model has learned during training, it assigns different probabilities to possible next tokens, conditioned on the prior context.⁴² We provide a high-level illustration of this process in Figure 1. At each generation step, a token is then selected from this distribution, which determines whether the model continues “Harry” with “Potter,” “Truman,” or something else.

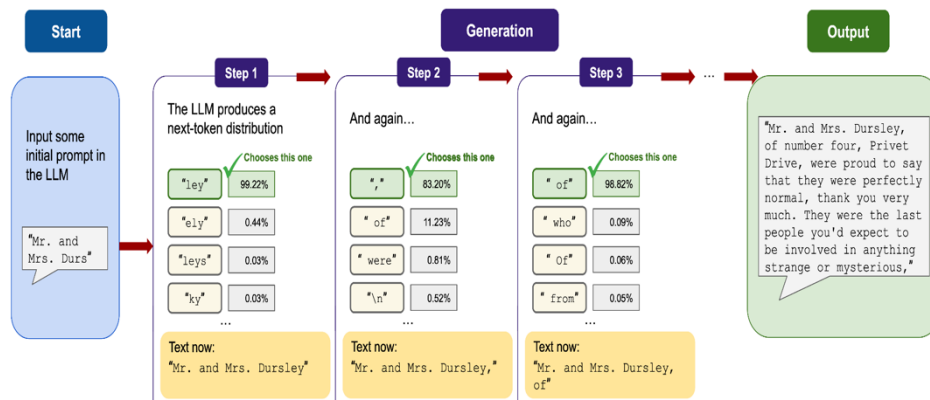


Figure 1: Illustrating autoregressive generation.

The input prompt is “Mr. and Mrs. Durs,” for which the (autoregressive) LLM produces a distribution over the next tokens in its vocabulary (all possible tokens it can generate). At each generation step, the decoding procedure illustrated here selects the highest-probability token as the one to generate. In this example, the top-ranked token has enormous probability at each step. The output after several steps is verbatim text from *Harry Potter and the Sorcerer’s Stone*. Figure produced by the authors, based on results for running this prompt on Llama 3.1 70B with greedy decoding.

This is where an important distinction arises between the model itself and the procedure used to generate outputs from it. The model defines a probability distribution over sequences. A given model is, in that sense, fixed once training is complete.⁴³ But generating text from that distribution requires a decoding

41. See Cooper & Grimmelmann, *supra* note 9, at 189. There are other types of language models, which produce outputs in different ways. Currently, autoregressive language models are the most common paradigm. See *supra* note XX. [Currently footnote 27]

42. See Cooper & Grimmelmann, *supra* note 9, at 192–93. Hannibal Travis calls LLMs a lexicon of words, grammar, and concepts. Hannibal Travis, *The Universal Lexicon: Generative AI Chatbots and Fair Use of Works in a Training Corpus*, 45 *Rev. Litig.* 60, 73 (2026).

43. It is fixed with respect to training being completed at the moment in time when the model is used. But the model could later be updated with additional training (e.g., fine-tuning), which would change its weights. So it’s more precise to say that the model is fixed with respect to the moment in

procedure — a rule for selecting tokens from the model’s predicted probabilities. In practice, decoding is often stochastic. Tokens are sampled from the probability distribution represented by the model, which means that the same prompt can yield different outputs on different runs. During generation, sometimes the model might produce “Potter” after “Harry,” and sometimes it might produce “Truman;” how often it produces either depends on the learned distribution in the model weights.

But decoding need not be stochastic: there are also deterministic decoding procedures. For instance, greedy decoding always selects the highest-probability token from the next-token distribution at each generation step. Because the learned probability distribution is fixed, at each generation step for a given prompt, that top-ranked token is the same every time. There is no randomness involved in selecting it, and the model produces the same output every time for the given prompt.⁴⁴

This distinction explains both the variability of model outputs and the limits of familiar analogies. Because the model itself defines a fixed probability distribution, the variability in outputs does *not* arise from changes in the model, but from how that distribution is used during decoding. The decoding procedure determines which of the (typically) many⁴⁵ possible continuations represented by the model are realized in any particular output.⁴⁶

Seen in this way, it becomes clear why analogies between models and traditional databases can be misleading. Language models don’t store responses in a discrete, localized way and then deterministically return those responses.⁴⁷

which it’s used, rather than to take this as a general statement about a model being permanently fixed. See Talkin’, *supra* note 20, at I.C.5.

44. See Cooper & Grimmelmann, *supra* note 9, at Part II.D; see generally A. Feder Cooper, Jonathan Frankle, and Christopher De Sa, *Non-Determinism and the Lawlessness of Machine Learning Code*, in Proceedings of the 2022 Symposium on Computer Science and Law (Association for Computing Machinery, 2022), <https://doi.org/10.1145/3511265.3550446>. This is true in principle, but may vary in practice. For instance, it’s true for a given loaded model run with a particular generation configuration on particular hardware. But loading the model weights in a different numerical precision, using generation with or without a KV cache, or other runtime differences may have subtle impacts on the distribution that change which token is ranked at the top. At the system level, there are additional considerations that impact non-determinism of model outputs. For instance, many deployed models are mixtures of experts (MoEs)— there are actually many models, as opposed to a single one. Different prompts get routed to different experts, and the routing decisions may depend on a variety of factors, introducing non-determinism. Nevertheless, for an open-weight model being used with a deterministic decoding algorithm on specific hardware, this is generally true. The results for extracting effectively a verbatim copy of *Harry Potter and the Sorcerer’s Stone* using Llama 3.1 70B are deterministic, for the different hardware we ran our code on: with the same prompt, you get the same output. See *infra* Part I.B, discussing Extracting Books (Open-Weight).

45. Memorization, as we will see soon, is an exception to this: for memorized sequences, there are *not* always “many possible continuations” for reasonable decoding procedures; rather, there may effectively only be *one* or a handful of possible continuation. See *infra* next section.

46. See Cooper & Grimmelmann, *supra* note 9, at XX.

47. *Id.* at 189–90.

Instead, they generate outputs from a probability distribution shaped by the training data, with decoding determining which continuation is actually produced in response to any given prompt. These two components – model weights and decoding procedures – together produce content in response to an initial prompt. This distinction is essential for understanding memorization and its relationship to model outputs.⁴⁸

One final clarification is worth noting. The discussion above concerns language models in isolation – that is, the model as defined by its weights and the probability distribution those weights induce over token sequences. This is a useful description for understanding open-weight models, such as Meta’s Llama models⁴⁹ and Alibaba’s Qwen models,⁵⁰ which can be downloaded and used directly with a user’s chosen decoding procedure.

In practice, however, many end users interact with language models that are embedded within larger proprietary systems: software services deployed as products like Claude or ChatGPT. These systems generally include additional layers, such as reinforcement learning, prompt processing or output filtering, that constrain or otherwise shape the outputs returned to users. As a result, for these systems, decoding is not the only mechanism that determines outputs; system-level features also play a significant role.⁵¹ Those features may mean that a model actually deployed to the public will (typically) not return a particular output to end users even though the model has memorized the content of that output.

B. Memorization in the Model, Extraction in Model Outputs

Following from above, we can now explain what the machine learning literature refers to as “memorization.”⁵² In the discussion that follows, a key takeaway is that, when it comes to memorization, it isn’t possible to make broad generalizations about AI, or even generalizations about a specific AI model with respect to all the training data it’s been trained on.

As detailed above, a trained language model defines a probability distribution over token sequences, where that distribution is learned from the training data. The technical notion of memorization can be understood as a

48. See *infra* I.B. Memorization is compressed information — it’s compressed storage — of specific training data in the model; extraction effectively reads those compressed stored data out of the model as output.

49. See generally Aaron Grattafiori et al., *The Llama 3 Herd of Models* (Nov. 23, 2024) (unpublished manuscript), <https://arxiv.org/abs/2407.21783>.

50. See generally An Yang et al., *Qwen2.5 Technical Report* (Jan. 3, 2025) (unpublished manuscript), <https://arxiv.org/abs/2412.15115>.

51. See Talkin’, *supra* note 20, at I.C.; see generally Machine Unlearning, *supra* note 3028. See *supra* note xx about non-determinism.

52. For a significantly more detailed treatment on this topic, we refer to Cooper & Grimmelmann, *supra* note 9. Here, we try to pare things down to what we consider the absolute minimum information needed to dispel misunderstandings of metaphors about “memorization” and “compression.” See *infra* Part I.C.

property of that distribution: loosely speaking, memorization is when, based on its training, the model places unusually high probability on specific sequences from its training data. Thus, those sequences are far more likely to appear in the output than alternatives. This doesn't mean the model ceases to be probabilistic; it still defines a probability distribution and may still give non-zero probability to many possible sequences. But the distribution is so sharply peaked that one sequence dominates.⁵³

This is what is meant, in a technical sense, when we and others in the machine learning community say that memorized training data are encoded in the model.⁵⁴ In that same technical sense, the point is closely related to compression, which helps make the claim more precise. As noted above, learning involves compression: a high-quality trained model stores a compact internal representation of the data that is sufficient for accurate prediction.⁵⁵ That compressed representation may capture broad patterns or abstractions, rather than preserving much that is specific to any particular training sequence. This is the intuition behind Ted Chiang's description of ChatGPT as a "blurry JPEG of all the text on the Web."⁵⁶ But memorization reveals the limits of that analogy: in certain cases, compression preserves particular training data with high fidelity. As Cooper & Grimmelmman write, it is perhaps best to

...refine Chiang's blurry JPEG compression analogy: not all parts of the JPEG are equally or completely blurry. Some training data are compressed more than others, and compression happens in different ways. Some of the compression is lossy: the information is discarded or transformed. But the portions that contain exact memorization are lossless [compression] and the

53. To understand "sharply peaked": imagine rolling a weighted die, where there's a 95% chance of rolling 3, and the remaining 5% is shared among 1, 2, 4, 5, and 6. This intuition, while useful, is also a simplification in several respects. Scientific research shows that memorized sequences need not be verbatim copies of training data, but may instead be very close variants (e.g., the memorized sequence replaces a comma with a semi-colon). In fact, many near-exact variants may be memorized simultaneously: a model may concentrate probability on a set of such near-verbatim sequences, for example, sequences that differ only in punctuation or exhibit minor syntactic variation. This is still memorization. More broadly, the scientific understanding of memorization continues to evolve. The account here is intended as a high-level description of the aspects that are relevant for copyright. See Daphne Ippolito et al., *Preventing Generation of Verbatim Memorization in Language Models Gives a False Sense of Privacy*, in Proceedings of the 16th International Language Generation Conference 28, 28 (Association for Computational Linguistics, 2023) (arguing that "verbatim memorization definitions are too restrictive and fail to capture more subtle forms of memorization"); See Near-Verbatim, *supra* note 13, at XX (developing new search models to understand near-verbatim extraction risk).

54. See Extracting Training Data from LLMs, *supra* note 153, at XX; Extracting Training Data from Diffusion Models, *supra* note YY, at XX; Nasr et al., *supra* note 12, at XX; Carlini Blog, *supra* note 15; Extracting Books (Open-Weight), *supra* note __, at XX; Near-Verbatim, *supra* note 13, at XX; Franceschelli, Cevenini & Musolesi, *supra* note 15, at XX; Cooper & Grimmelmman, *supra* note 9, at XX.

55. See Delétang et al., *supra* note __, at XX; Mostaque First Amended Complaint at 27, Andersen v. Stability AI, Ltd., No. 3:23-cv-00201 (N.D. Cal. Nov. 29, 2023).

56. Ted Chiang, *ChatGPT is a Blurry JPEG of the Web*, NEW YORKER, Feb. 9, 2023.

portions that contain near-exact memorization are nearly lossless [compression.]⁵⁷

So far, we haven't said *anything* about model outputs; what we've just described – memorization – is a property of the model itself. This property is not directly observable. As discussed in Part I.A, the internal structure of models is difficult to interpret, and so we typically can't identify where or how particular information is stored, including memorized training data. As a result, memorization in generative AI is usually studied through the outputs that a model produces under carefully chosen conditions. This isn't because outputs define memorization, but because they provide a way of observing a property of the model that would otherwise remain hidden.

Figure 2: Four portions of the text of Harry Potter and the Sorcerer's Stone, generated by the authors using Llama 3.1 70B.

He came ter yer house an'—¹ **just—** ² **ter—** ³ **ter—** ⁴ **ter—** ⁵ **ter—** ⁶ **ter—** ⁷ **ter—** ⁸ **ter—** ⁹ **ter—** ¹⁰ **ter—** ¹¹ **ter—** ¹² **ter—** ¹³ **ter—** ¹⁴ **ter—** ¹⁵ **ter—** ¹⁶ **ter—** ¹⁷ **ter—** ¹⁸ **ter—** ¹⁹ **ter—** ²⁰ **ter—** ²¹ **ter—** ²² **ter—** ²³ **ter—** ²⁴ **ter—** ²⁵ **ter—** ²⁶ **ter—** ²⁷ **ter—** ²⁸ **ter—** ²⁹ **ter—** ³⁰ **ter—** ³¹ **ter—** ³² **ter—** ³³ **ter—** ³⁴ **ter—** ³⁵ **ter—** ³⁶ **ter—** ³⁷ **ter—** ³⁸ **ter—** ³⁹ **ter—** ⁴⁰ **ter—** ⁴¹ **ter—** ⁴² **ter—** ⁴³ **ter—** ⁴⁴ **ter—** ⁴⁵ **ter—** ⁴⁶ **ter—** ⁴⁷ **ter—** ⁴⁸ **ter—** ⁴⁹ **ter—** ⁵⁰ **ter—** ⁵¹ **ter—** ⁵² **ter—** ⁵³ **ter—** ⁵⁴ **ter—** ⁵⁵ **ter—** ⁵⁶ **ter—** ⁵⁷ **ter—** ⁵⁸ **ter—** ⁵⁹ **ter—** ⁶⁰ **ter—** ⁶¹ **ter—** ⁶² **ter—** ⁶³ **ter—** ⁶⁴ **ter—** ⁶⁵ **ter—** ⁶⁶ **ter—** ⁶⁷ **ter—** ⁶⁸ **ter—** ⁶⁹ **ter—** ⁷⁰ **ter—** ⁷¹ **ter—** ⁷² **ter—** ⁷³ **ter—** ⁷⁴ **ter—** ⁷⁵ **ter—** ⁷⁶ **ter—** ⁷⁷ **ter—** ⁷⁸ **ter—** ⁷⁹ **ter—** ⁸⁰ **ter—** ⁸¹ **ter—** ⁸² 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that these models have memorized significantly more text from books in their training data than was previously understood. In an extreme instance, prompting Meta’s Llama 3.1 70B model with just a short snippet from the opening of *Harry Potter and the Sorcerer’s Stone* was sufficient to generate a near-exact copy of the entire 304-page book. For behavior like this, the only reasonable explanation is memorization of the book from the training data.⁶⁰ Figure 2 shows what we mean by near-exact: for four portions of the generated text, it shows the differences between the generated text and the reference, ground-truth text we have available from the Books3 corpus.⁶¹ Using the same code with Llama 3.1 70B and a prompt of only the first sentence, we also generated over 10,000 words near-verbatim from “A Case for Reparations” (included in *We Were Eight Years in Power*), an essay written by Ta-Nehisi Coates, who was one of the *Kadrey v. Meta* class-action plaintiffs.⁶²

This behavior can be understood in terms of the probability distribution defined by the model. Starting from this very short prompt, the next tokens in the original text have overwhelmingly higher probability than plausible alternatives. Figure 1 provides a simplified illustration for *Harry Potter and the Sorcerer’s Stone*, showing the prompt, the model’s highest-probability tokens at each step, and the token actually selected during decoding. The ground-truth text from the book consistently has by far the highest probability. The distribution is so sharply peaked that, in practice, the generated output tracks the memorized text. For Llama 3.1 70B, this is true throughout *Harry Potter and the Sorcerer’s Stone*,⁶³ such that it is possible to deterministically reproduce the entire book. This type of behavior – the generation of memorized training data in model outputs – is commonly referred to in the machine learning literature as “extraction.”⁶⁴

60. For an exhaustive treatment of this, see Extracting Books (Open-Weight), *supra* note 12, at 163–181; see also Carlini Blog, *supra* note 15; *supra* compression paragraph; Cooper & Grimmelmann, *supra* note 9, at XX.

61. The Books3 corpus contains approximately 200,000 torrented ebooks, converted to text-format for the purpose of LLM training. Alex Reisner, *Revealed: The Authors Whose Pirated Books are Powering Generative AI*, The Atlantic, Aug. 19, 2023; Talkin’, *supra* note 20, at XX; see generally Katherine Lee et al., *AI and Law: The Next Generation* (Oct. 19, 2023) (unpublished manuscript), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4580739.

62. Extracting Books (Open-Weight), *supra* note 12, at 72. *We Were Eight Years in Power* was one of the books named in the lawsuit.

63. The probabilities are so high that we are able to deterministically decode the whole book using a decoding algorithm called beam search. See Extracting Books (Open-Weight), *supra* note 12, at XX.

64. See Cooper & Grimmelmann, *supra* note 9, at XX, (giving a more precise taxonomy of terms and unpacking some distinctions in the literature). Extraction is intentional prompting to attempt to reproduce training data in outputs. Regurgitation is unintentional – when this happens, regardless of trying to do this. The frequency of regurgitation in language models is poorly understood and not often studied in the literature. There are two exceptions (only sort of) in very limited and controlled research settings. See generally Michael Aerni et al., *Measuring Non-Adversarial Reproduction of Training Data in Large Language Models*, in The Thirteenth International Conference on Learning Representations

Figure 3: Running the same long-form extraction procedure on *Sandman Slim*. The blue text is the prompt we provided, which is the first few sentences of Richard Kadrey's book. The black text is the generation. We tried prompts of various lengths, but produced no generations that were near-verbatim reproductions of text in the ground-truth book. This experiment doesn't show successful extraction of memorized training data. (Interestingly, though, the generation does contain references to characters in the book, which we did not provide in the prompt:

James Stark, Sandman Slim, etc.)

I WAKE UP on a pile of smoldering garbage and leaves in the old Hollywood Forever cemetery behind the Paramount Studio lot on Melrose, though these last details don't come to me until later. Right now all I know is that I'm back in the world and I'm on fire. My mind hasn't quite kicked in yet, but my body knows enough to roll off the burning trash and to keep rolling until I can't feel the heat anymore. I come to a stop against a gravestone and lie there for a moment, trying to remember how I got here and who I am. I don't always know right away. It's the amnesia. It's a hell of a way to travel, but it's better than being dead. Slowly, like dirty water dropping out of a rusty pipe, it comes back to me. My name is James Stark. Friends call me Sandman Slim. Call me what you like. I've been called worse. I got this name because all I do is bring nightmares. True, I spend a lot of time in the City of Angels, but it's not like they're putting up a statue to me. Before this, I lived in Hell for eleven years. Even when I tried to stay out of trouble, trouble would come right to my door and make friends. That kind of thing has a way of making a man a little jaded when it comes to trusting his fellow man. Or demons. Even angels. I'm like that. I've got a bad attitude. I drink. I've been kicked out of every twelve-step program I've ever joined. I don't have many friends, so the ones I have mean a lot to me. They mean enough that if someone hurts them, I'll tear the universe down to save them. I'm telling you this so you'll know who I am and what I'm talking about when I say there are things in the dark. Things we don't always see that move among us. Some of these things are good and some aren't and sometimes they do things that make you want to rip open your chest and pull out your heart because you've seen too much and there's no going back. I know about these things because I'm one of them. I'm a monster and I'm a killer. I'm good at it and I never let emotion get in the way. When you live where I live, love can get you killed. And when it doesn't, it's worse because if you don't die, you just wish

Extraction isn't always possible. In typical cases, a model generates text that doesn't reproduce memorized training data. As a counterpoint to the *Harry Potter* example, consider our attempt to apply a similar procedure for another Kadrey v. Meta book, Richard Kadrey's *Sandman Slim*, which we illustrate in Figure 3. We know that this book was in Llama's training data,⁶⁵ so it's at least possible in principle for the model to have memorized text from it, just as it memorized the first *Harry Potter* book or Ta-Nehisi Coates' essay. But in practice, using the same procedure, the model doesn't reproduce the book. Instead, it produces a range of plausible outputs, reflecting a distribution that isn't concentrated on any single sequence, let alone the ground-truth text from the training data.

These contrasting examples illustrate an important point: memorization isn't a uniform phenomenon that applies equally across a model's training data. Some training data are memorized, while others aren't. Extraction is an imperfect tool for measuring memorization:⁶⁶ it can, by definition, only produce sequences

(ICLR, 2025), <https://openreview.net/forum?id=590yfqz1LE>; Parjanya Prajakta Prashant, Kaustubh Ponshe & Babak Salimi, *TokenSwap: A Lightweight Method to Disrupt Memorized Sequences in LLMs*, in The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS, 2026), <https://openreview.net/forum?id=gNiT8liag0>. We defer to Cooper & Grimmelmann for more precise treatment, but will call this extraction here.

65. Kadrey v. Meta Platforms, Inc., 788 F. Supp. 3d 1026, 1042 (N.D. Cal. 2025).

66. Specific extraction procedures are selective in additional ways. For instance, a given procedure may test only a certain set of prompts, and miss identifying memorization that other prompts would have revealed. Research using greedy decoding misses large amounts of extraction that ours and

that have been memorized,⁶⁷ and not all memorized training data are necessarily extractable.⁶⁸ In this second sense, extraction is an imperfect measurement tool: it provides evidence of memorization where it succeeds, but its failure doesn't necessarily imply that memorization hasn't occurred. Memorization isn't a Schrödinger's-cat-like situation: memorized training data do not both exist and not exist depending on whether they're observed.. Training data may or may not be encoded in the model; and if they are, they are there regardless of whether they can be extracted and whether we can prove it.⁶⁹

Further, models don't all compress information about their training data in the same way. Sometimes they capture high-level relationships across data points without memorizing specific text sequences, and sometimes they encode specific sequences (i.e., memorize them). Sometimes memorized training data can be extracted deterministically, and sometimes only probabilistically, with varying frequency.⁷⁰ These details — the probabilistic elements, how compression works, etc. — are the core of the issue and can't be abstracted away.⁷¹

But even with this clarifying perspective, much about memorization remains poorly understood. Some factors, such as duplication of sequences in training data, are known to increase the likelihood of memorization.⁷² However, in general, there is no single answer to how much memorization occurs, or precisely when and why it arises, in part because our tools for measuring it are imprecise at best. For instance, some prior estimates suggest that somewhere

other work has identified using probabilistic extraction, and so on. See "Extracting Books (Open-Weight)". First citation to probabilistic extraction should go here [see footnote 73, Hayes et al.]

67. An important consideration here is that the measurement procedure for extraction has to be *valid*. If it isn't valid, then the "extraction" procedure could produce outputs that are not actually extraction, but chance generation of *non*-training data that would erroneously get interpreted as evidence of memorization. For rigorous scientific work on extraction and memorization, this is not an issue. Nevertheless, not all published work in the technical literature meets this bar. The invalidity of some work that calls itself "extraction" does not cut against the validity of work that rigorously studies extraction. However, it's important to be vigilant about this when assessing extraction claims. This of course isn't unique to extraction; distinguishing between valid and invalid measurement methodologies and resulting claims is a challenge for empirical sciences, more generally. *See generally* A. Feder Cooper et al., "Extractable Memorization From First Principles" (2026) (unpublished manuscript).

68. Cooper & Grimmelmann, *supra* note 9, at XX; Nasr et al., *supra* note 12, at 1 (distinguishing discoverable and extractable memorization). *See supra* note 70 (the new paper cited in that footnote).

69. *See infra* Part I.C for a discussion of how this point is frequently misunderstood: extraction doesn't determine memorization (this flips the relevant direction); memorization is a prerequisite for extraction.

70. Jamie Hayes et al., *Measuring Memorization in Language Models via Probabilistic Extraction*, in Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies Volume 1: Long Papers (Association for Computational Linguistics, 2025), <https://aclanthology.org/2025.naacl-long.469/>; Near-Verbatim, *supra* note 13, at XX.

71. *See* Cooper & Grimmelmann, *supra* note 9, at 197 ("[T]he probabilistic element . . . is inescapable.")

72. *See generally* Deduplicating, *supra* note 13.

between 0.5% and 10% of training data sequences may be memorized by language models.⁷³ While these numbers might sound like firm, general, authoritative claims, they are in fact based on very specific experimental setups.⁷⁴ They are average estimates computed using specific extraction procedures on particular models and datasets.

Changing any of these factors – the extraction method, the model, or the training data of interest – can significantly affect the results. Much of the early work relied on greedy decoding to measure extraction, which selects the highest-probability token at each decoding step.⁷⁵ More recent approaches allow for more realistic non-deterministic decoding schemes and repeated sampling from the same prompt, which can reveal additional instances of memorization that greedy methods miss.⁷⁶ In our own work, we quantify how running the same prompt hundreds or thousands of times can produce verbatim extraction of copyrighted text in some runs,⁷⁷ but for many works, a single verbatim extraction of fifty tokens from a work might be accompanied by hundreds of outputs that are not verbatim extractions.⁷⁸ And our more recent work shows that even these newer approaches undercount the extent of memorization, because many of the “failures” to extract a sequence in fact generate near-verbatim copies of that sequence.⁷⁹

73. See Cooper & Grimmelmann, *supra* note 9, at XX; Extracting Training Data, *supra* note 15, at XX; Nasr et al., *supra* note 12, at 15 (“[T]his means that approximately 10% of the entire model capacity is “wasted” on verbatim memorized training data.”)

74. For example, Morris et al. make an aggregate claim that models in the GPT family have a 3.6 bits-per-parameter memorization capacity. John X. Morris et al., How Much Do Language Models Memorize? XX (June 18, 2025) (unpublished manuscript), <https://arxiv.org/abs/2505.24832>. This result is derived with respect to a very specific experimental training conditions for small models – a setup that is significantly different from those used to train production LLMs. *Id.* at XX. So while this work offers a precise scientific claim, in isolation, it is easy to misconstrue the “3.6 bits” number as a general claim about LLMs, as in this news coverage of the scientific paper: Carl Franzen, *How Much Information Do LLMs Really Memorize? Now We Know, Thanks to Meta, Google, Nvidia and Cornell*, VentureBeat, June 5, 2025. See generally Cooper et al. (2026), *supra* note 70 [the new paper that doesn’t yet have an arXiv link] (concerning how the practical setup for extraction experiments influences conclusions about memorization).

75. See *supra*, Part I.A (defining greedy decoding). See Deduplicating, *supra* note 1372, at XX; Gemini Team Google et al. Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context XX (Dec. 16, 2024) (unpublished manuscript), <https://arxiv.org/abs/2403.05530>; Aaron Grattafiori et al., *supra* note 4947, at XX; Gemma Team et al., *Gemma 2: Improving Open Language Models at a Practical Size* XX (Oct. 2, 2024) (unpublished manuscript), <https://arxiv.org/abs/2408.00118>.

76. Hayes et al., *supra* note 7068, at XX.

77. This work validates that it’s actually capturing extraction, as opposed to chance generation that resembles training data. We defer the details to that work, but emphasize here that generating training data one out of every 1000 attempts with the same prompt is strong evidence for memorization in the work we conduct. See Extracting Books (Open-Weight), *supra* note 12, at XX. See also Cooper (2026), *supra* note 70 [Extractable Memorization From First Principles, new paper with pending arXiv link]

78. See Extracting Books (Open-Weight), *supra* note 12, at XX.

79. Near-Verbatim, *supra* note 13, at XX; Ippolito, *supra* note 5354, at XX.

What we *can* say with some confidence from the empirical extraction literature is that memorization doesn't reduce to a simple story. It's not correct to conclude either that AI models always memorize specific content or that they almost never do. Instead, the extent and specificity of memorization varies significantly across models (e.g., Llama versus Qwen), across versions within the same model family (e.g., Llama 2 versus Llama 3), across model sizes (e.g., Llama 2 7B versus Llama 2 70B), and across individual works for a given model.⁸⁰ And for models that are part of deployed AI systems, additional system-level factors further affect whether memorized training data are observed in outputs.⁸¹ As recent work shows, it's definitely possible to extract some copyrighted training data, but, depending on the system, this can require different methods altogether.⁸² So, in practice, when it comes to memorization it isn't possible to make broad generalizations about AI, or even generalizations about a specific AI model with respect to all the training data it's been trained on.

C. Metaphors, Misunderstandings, and the Problem of Dismissing Technical Precision

Having gone beneath the metaphorical surface of terms like memorization and compression, it should now be clearer why those terms are not exhausted by their lay interpretations.⁸³ In machine learning, they name technical phenomena. The mistake, then, is not merely to notice that the metaphors are imperfect — they very much are. It is to treat that imperfection as a reason to discount or abstract away the technical particulars.

A model is not literally arranged like a filesystem or a database. But that doesn't really matter,⁸⁴ and that also doesn't mean it stores nothing.⁸⁵ It stores

80. Nicholas Carlini et al., *Quantifying Memorization Across Neural Language Models* at 1, INT'L CONF. LEARNING REPRESENTATIONS (2023); Hayes et al., *supra* note 7068, at XX; Near-Verbatim, *supra* note 13, at XX (probabilistic extraction and near-verbatim probabilistic extraction, for the same point); *see also* Extracting Books, *supra* note __, at XX (for the books-specific variation in memorization claims, across books, models, model sizes, and versions).

81. *See* Cooper & Grimmelmann, *supra* note 9, at Part II.I (discussing system-level modifications that can prevent memorized training data from being produced to the user even if it is generated by the model); Machine Unlearning, *supra* note 3028, at 5.

82. *See* Nasr et al., *supra* note 12, at 9 (“divergence” jailbreak of asking the model to repeat the same word forever, e.g. “poem,” resulted in some outputs leaking verbatim training data from ChatGPT 3.5). To extract copyrighted books from proprietary models, Ahmed et al. used jailbreaks to elicit memorization from ChatGPT 4.1 and Claude 3.7 Sonnet, but did not need to jailbreak Grok 3 or Gemini 2.5 Pro to elicit enormous amounts of copyrighted material. In all cases, they were able to bypass content filters to reproduce copyrighted material in system outputs). *See* Extracting Books (Production LLMs), *supra* note 13, at XX

83. *See supra* Part I.A

84. *See infra* II.A, current footnote 100ish about copies and computer encodings in binary

85. *See* Aline Larroyed, *The Fallacy of the File: How the Memorisation Metaphor Misguides Copyright Law and Stifles AI Innovation* XX (Nov. 21, 2025) (unpublished manuscript), <https://ssrn.com/abstract=5782882> (conflating the technical details of memorization with the

information, in bits, in a technically meaningful sense.⁸⁶ In the case of memorized training data that can be extracted, information about those data is encoded in the model's weights with enough fidelity to make those data recoverable in outputs. Extraction is important for precisely that reason: it's evidence of training data that are encoded in the model's weights; it "is a symptom of memorization in the model, not its cause."⁸⁷ One therefore can't take extraction seriously while setting aside the memorization in the model, because the former is only possible because of the latter.

It's also true that training a model isn't literally like creating a ZIP file or converting a song into MP3 format, and so thinking about learning and compression in those terms can be misleading. But that observation answers only the everyday analogy and the possible misuse courts or litigants might make of it; it doesn't address the underlying technical claim.⁸⁸ In cases of memorization, compression preserves certain specific training data with high fidelity. For that reason, compression helps explain what has been retained in the model and why

metaphorical aspects of the term memorization; taking the metaphor in the title of Cooper & Grimmelmann's "*The Files Are in the Computer: On Copyright, Memorization, and Generative AI*" too literally, taking the term "memorization" only on metaphorical rather than technical terms, and then arguing that because memorization is both metaphorical and not exactly like file storage, it's irrelevant to copyright).

86. See generally Delétang et al. *supra* note XX. . See generally Zhiying Jiang et al., "Low Resource" Text Classification: A Parameter-Free Classification Method with Compressors XX (July 2023) (unpublished manuscript), <https://aclanthology.org/2023.findings-acl.426/>. Thomas M. Cover and Joy A. Thomas, *Data Compression*, 103–158 (John Wiley & Sons, Ltd, 2005).

87. Cooper & Grimmelmann, *supra* note 9, at 142. See *supra* notes __, __ for this point. See also Adi Haviv et al., *We Should Separate Memorization from Copyright* XX (Feb. 9, 2026) (unpublished manuscript), <https://arxiv.org/abs/2602.08632>. Haviv et al. emphasize that model weights aren't directly interpretable and that training data aren't stored in a database-like or file-like format. Then, from that basis, they cast extraction as too mediated by prompts and prior knowledge to be evidence of memorization encoded in the model itself. They explain that point by analogy to fMRI reconstructions: brain signals may contain information about an image, but the recognizable image emerges only after a separate reconstruction step infers an image-like output from those signals. But that analogy reverses a basic causal point. Prompts and prior knowledge bear on the conditions under which memorized training data becomes observable, not on whether those training data were encoded in the model in the first place. A selective or prompt-dependent extraction result shows only that memorized information isn't uniformly decodable from the model's weights; it doesn't show that the information was supplied by the extraction procedure rather than encoded through training. Indeed, properly validated extraction procedures are designed to rule out the prompt itself as the source of the recovered material. See *Extracting Books (Open-Weight)*, *supra* note 12, at XX; Cooper *Untitled*, *supra* note 3129, at XX. The fMRI comparison is therefore misleading here. An fMRI scanner produces only an indirect physiological trace, and the recognizable image appears only after a separate system maps that trace back into the image space. By contrast, in extraction, there's no separate reconstruction system generating the recovered content: the model being studied generates the sequence itself.

88. Sag, *supra* note 4038, at XX. (recognizing that there are instances of memorization – the technical model behavior – that are salient to copyright law; but rejecting the "fallacy of compression" on analogical grounds because "compression" invokes things like MP3 files in ordinary meaning. In doing so, the paper doesn't directly engage the technical aspects of compression and memorization that it simultaneously treats as relevant to copyright.)

some memorized training data may later be more readily recoverable than others.⁸⁹

The point, then, is not to discard terms like memorization and compression because their everyday meanings are imperfect guides to how generative AI works. It's to engage them on their technical particulars, as we have done here, because doing so reveals details about what's in the model and what may be recovered in outputs — details that matter for legal analysis. Part II turns to those consequences for copyright.

II. THE LEGAL COMPLICATIONS OF A “COPY”

Part I showed that memorization presents several technical uncertainties. New extraction methods continue to reveal memorized material that earlier methods may not expose, making it difficult to say with confidence that a particular piece of training data definitively hasn't been memorized by a particular model. And where extraction is possible, in some cases it may occur only probabilistically, rather than being deterministically guaranteed. To these technical uncertainties, we now add a discussion of legal uncertainty.

It's unclear how the definition of a “copy” applies to the storage of information that can sometimes be used probabilistically to generate identical or substantially similar outputs in response to a prompt. As discussed below, we can't simply rely on the statutory definition of a copy, in part because it is itself internally self-contradictory. And while courts have applied it to a variety of situations, they haven't confronted one quite like this before. How the courts decide to treat stored model weights that can be used to probabilistically (and, in some cases, deterministically⁹⁰) generate different outputs, some but not all of which correspond to prior works, will have significant consequences for the potential liability of AI companies, the viability of open-source business models, and the feasibility of efforts to minimize output infringement and of licensing efforts.

A. *The Statute Doesn't Tell Us Whether Model Weights Are a Copy*

Section 101 of the Copyright Act is normally quite a good guide to the meaning of terms. Unfortunately, that's not true here.

We begin with a significant problem, albeit one tangential to our issue: the definition of copy in the act is self-contradictory.⁹¹ Read literally, nothing can

89. Hayes et al., *supra* note 7068, at XX; Near-Verbatim, *supra* note 13, at XX; Cooper Untitled, *supra* note 3129, at XX.

90. See *supra* re: greedy decoding; Harry Potter result for Llama 3.1 70B.

91. This hasn't really been discussed in the prior literature. Tony Reese and Tyler Ochoa advert to the confusing language in footnotes. See R. Anthony Reese, *Public Display Right*, 2001 U. ILL. L. REV. 83, 127 n. 171; Tyler Ochoa, *Copyright, Derivative Works and Fixation: Is Galoob a Mirage, or*

possibly be an infringing copy. Section 106 says “the owner of copyright under this title has the exclusive rights to do and to authorize any of the following: (1) to reproduce the copyrighted work in copies.”⁹² So, not surprisingly, one infringes a copyright by making a copy. Section 101 defines “Copies” as “material objects, other than phonorecords, in which a work is fixed.”⁹³ That means that an infringing copy doesn’t include transitory things like projection of a movie on a screen or data that transits a wire or radio waves in electronic form. But section 101 also says that “a work is ‘fixed’ in a tangible medium of expression when its embodiment in a copy or phonorecord, *by or under the authority of the author*, is sufficiently permanent or stable to permit it to be perceived, reproduced, or otherwise communicated for a period of more than transitory duration.”⁹⁴

Putting those three together and taking only the literal language seriously, the only potentially infringing “copy” is one that is made “by or under the authority of the author” – and therefore is presumably not infringing.

To be clear, the point here is not that a court is likely to conclude that nothing is copyright infringement. Courts have consistently ignored (or perhaps not discovered) this problem in the statute and proceeded to declare infringing acts copies.⁹⁵ Rather, our point is that because the statute is incoherent on its face, we ultimately can’t rely on plain meaning in interpreting the statute, and may need to turn to policy, logic, and the purpose of the law to understand it.⁹⁶

At a minimum, though, the statute tells us that to be a copy, the copyrighted work in the model must be fixed in a material object. Computers definitely count as material objects,⁹⁷ but there must be actual storage of the copyrighted work in the computer, not just the potential for the computer to generate the work.

That storage need not look like the object in the real world. It’s well established that a computer that stores a book or a movie in the form of electrical impulses that translate to 1s and 0s still contains a copy. The definition of a copy addresses this directly; a copy is a material object “in which a work is fixed by any method now known or later developed, and from which the work can be perceived, reproduced, or otherwise communicated, either directly or with the aid of a machine or device.”⁹⁸ So it’s certainly possible that a work could be

Does the Form (GEN) of the Alleged Derivative Work Matter?, 20 Santa Clara Comp. & High Tech. L.J. 991, 997 n. 26 (2004).

92. 17 U.S.C. § 106.

93. 17 U.S.C. § 101.

94. *Id.*

⁹⁵ The problem arises from the perhaps unfortunate choice to define “copy” as central to both copyrightability and infringement without distinguishing the two. For a discussion, see Paul Goldstein, Goldstein on Copyright 7:14-7:14.1.

⁹⁶ It also is yet another piece in the mountain of evidence that “plain meaning” is a rather foolish doctrine to adhere to blindly. But that’s beyond the scope of the paper.

⁹⁷ Congress amended the Copyright Act in 1980 to add a definition of computer programs, and the legislative history indicates an intent that a computer program be treated as a literary work, but curiously the amended statute nowhere says as much.

98. 17 U.S.C. § 101.

“fixed” in a series of model weights – *if* it can reliably be perceived or reproduced from those model weights.⁹⁹

B. The Challenge of Non-deterministic Instantiations

The embodiment of the work must actually exist in the model in some semi-permanent form. The statute says the embodiment in a material object must be “sufficiently permanent or stable to permit it to be perceived, reproduced, or otherwise communicated.”¹⁰⁰ That turns out to be a key question for a non-deterministic technology like generative AI. In many prior technologies, the embodiment in a computer was encoded, but often deterministic: the same compiled code would create the same result by calling on the same 1s and 0s each time. That deterministic reproduction is what people had in mind when talking about fixation in a computer in the 1976 Copyright Act.¹⁰¹ If we think the AI is calling on a stored version of the work as a whole, that is clearly satisfied. (The scientific work in machine learning and extraction is consistent with this: there are bits stored in the model that encode memorized works, even if current scientific methods can’t localize which model weights reflect that encoding.) If, however, we think the model is generating an output on the fly, it wouldn’t be; the user is fixing – and therefore creating – the output by prompting the AI, which creates something new in response to the prompt.¹⁰²

There are some prior technologies that sit somewhere in between those examples. One is “lossy compression.” While some computer programs store information with perfect fidelity in a database, often that takes up too much memory. Compression technologies allow a computer to store a smaller subset of information corresponding to, say, a music file or a photograph and then

99. The definition of a computer program might also seem relevant here. A computer program is “a set of statements or instructions to be used directly or indirectly in a computer in order to bring about a certain result.” 17 U.S.C. § 101. While that would seem to encompass model weights that generate a copyrighted output, the fact that the definition is nowhere connected to either copyrightability or infringement makes it less useful as a guide. *See generally* Cooper & Grimmelmann, *supra* note 9.

100. 17 U.S.C. § 101.

101. *See* National Comm’n on New Technological Uses of Copyrighted Works, Final Report at 44 (1979) (“In every case, the work produced will result from the contents of the database, the instructions indirectly provided in the program, and the direct discretionary intervention of a human involved in the process.”). CONTU rejected the idea that the computer itself could generate new content as “too speculative to require serious consideration.” Pamela Samuelson, *Allocating Ownership Rights in Computer-Generated Works*, 47 U. PITT. L. REV. 1185, 1195 (1986) (citing CONTU Report) [hereinafter “Allocating Rights”]. Samuelson notes that even in the 1980s that was incorrect in some cases: “Although both the programmer and user might contribute to the framework within which the computer makes its selections or arrangements of data, the computer actually makes the selection.” *Id.* at 1199, 1201.

102. *Id.* at 1202 (“the user will have been the instrument of fixation for the work, that is, the person who most immediately caused the work to be brought into being.”).

reconstitute the full song or image from the compressed data.¹⁰³ Some compression is “lossless:” the computer essentially has a shorthand way of storing a reference to all the information in the original file, and it expands that shorthand into the full information set on demand. Other compression is “lossy:” we sacrifice some image quality and fidelity in order to store the image in a smaller amount of space. A file saved with lossy compression doesn’t correspond exactly to the original work. Nonetheless, we would say a compressed file stored in a computer is a (partial) copy of the original work.¹⁰⁴ But the file, while an imperfect copy, is still deterministic, not probabilistic: the same file will generate the same imperfect copy every time.¹⁰⁵ A given model stored on a computer is also fixed; its weights don’t change from one generation to the next. But because those weights encode probabilistic relationships, even where a model has memorized training data, extraction may in some cases occur only probabilistically: the same prompt may sometimes elicit the memorized material and sometimes not.

U.S. copyright law hasn’t really had to confront whether copyrighted material that can be elicited from a computer system only probabilistically – appearing in some outputs but not others depending on the probabilities of each successive word appearing – is “fixed” in the sense required to qualify as a copy.¹⁰⁶ A computer program compiled from source code into object code (the

103. For a discussion of lossy compression in the AI context, see *See* Cooper & Grimmelmann, *supra* note 9, at 188–89. See *supra* Part I.B.

104. Many familiar file formats, like JPEG for images, are lossy. As a matter of information theory, it is an interesting problem how much information must be provided to “contain” a work. But this is out of our present scope. See Sarah Scheffler, Eran Tromer & Mayank Varia, *Formalizing Human Ingenuity: A Quantitative Framework for Copyright Law’s Substantial Similarity*, in *Proceedings of the 2022 Symposium on Computer Science and Law* 37–49 (Association for Computing Machinery, 2022), <https://doi.org/10.1145/3511265.3550444>.

105. Language modeling is compression in a precise information-theoretic sense: a probabilistic language model, paired with an appropriate coding scheme, induces a lossless compressor. This does *not* mean the model is a lossless archive of its training data. Rather, the claim is *distributional*: data drawn from the kind of distribution the model captures can be efficiently encoded *on average*. The underlying intuition is simple: predictable material takes less information to describe, and is therefore more compressible. So the better the model is at predicting what comes next, the fewer bits are needed to encode the resulting text, on average. In this sense, an LLM is an effective compressor because of its predictive capabilities. Importantly, this is *not* a claim that every piece of training data is encoded in a lossless or retrievable form, such that it can be reconstituted verbatim or near-verbatim in generation. Rather, verbatim and near-verbatim extraction reflects a specific, high-fidelity form of compression that is a hallmark of memorization, not of language modeling more generally. See *supra* I.B. See Gregorire Deletang et al., *Language Modeling Is Compression* at XX, Int’l Conf. Learning Representations (2024); Chiang, *supra* note __; Getty Images v. Stability AI, [2025] EWHC 2863 (Ch.) ¶ 555 (“[t]he model weights do not directly store the pixel values associated with billions of training images . . . A single training image has only a limited impact on how the parameters of the network will be changed.” See Cooper *Untitled*, *supra* note 3129, at XX.

106. A Munich court decision found that memorization did create a copy under EU copyright law, see *GEMA v. OpenAI*, 42 O 14139/24 (LG Munchen 11 Nov. 2025), though that decision is currently under appeal. See, e.g., Zbigniew Okon, *Memorization of Copyrighted Works by AI Models Under EU Law*, 12 *Europejski Przegląd Sądowy* 19 (2025), English translation at <https://download.ssrn.com/2026/1/8/6041395.pdf>; M. Leistner and L. Antoine, *TDM and AI Training*

1s and 0s that actually operate the program) doesn't look anything like the source code that generated it. But the relationship between the two is designed to be deterministic: the object code is the predictable result of compiling that source code with a particular compiler. That determinacy is what makes the compiled program reliably produce the behavior one would expect from the written source code. Even versions that require more judgment, like language translations, are nonetheless instantiated in a permanent form: once made, the relationship between the original and the translation doesn't vary from one access to the next.

The closest cases to present issues of non-determinism involve gardens and video games. In *Kelley v. Chicago Park District*,¹⁰⁷ the Seventh Circuit held that a garden was not copyrightable because it wasn't fixed, though a blueprint design for a garden could be.¹⁰⁸ The court found that "a garden is simply too changeable to satisfy the primary purpose of fixation; its appearance is too inherently variable to supply a baseline for determining questions of copyright creation and infringement."¹⁰⁹ The fact that gardens grow and change organically meant to the court that the way in which it would evolve was unpredictable.¹¹⁰

We think *Kelley* goes too far. While gardens certainly do grow and change in ways that are not fully predictable, both the initial conditions and the way the gardens are curated will significantly affect those changes in a way that seems likely to be mostly if not entirely reproducible. Certainly, the fact that people plan gardens with blueprints suggests that they have a design that, if not fully deterministic, is implementable and sufficiently replicable to be usable.

In some prior cases we see a limited form of non-deterministic behavior in the output of a computer program. In video games, for example, the game code generates images that interact with human behavior to produce gameplay sequences on the screen that are not themselves exactly stored in the computer. My experience of a combat sequence, for example, will differ from yours, because we won't hit the same keys in the same order. Nonetheless, we treat those audiovisual sequences as copies of the underlying work despite the interactivity of the game, which means that the precise sequence of images that

in the European Union — From "LAION" to Possible Ways Ahead?, 11 GRUR Int'l 1032 (2025). By contrast, an English court, evaluating a different model and modality (images) with different evidence presented, has held the opposite. See *Getty Images v. Stability AI*, [2025] EWHC 2863 (Ch.) ¶ 603.

107. 635 F.3d 290 (7th Cir. 2011).

108. *Id.* at 304–05. *Kelley* treats fixation as a matter of copyrightability, not infringement. But since the standards are the same and rely on the same statutory definition, presumably it would apply to the fixation required for infringement, as well.

109. *Id.*

110. *Id.* at 305 ("Although the planting material is tangible and can be perceived for more than a transitory duration, it is not stable or permanent enough to be called "fixed." Seeds and plants in a garden are naturally in a state of perpetual change; they germinate, grow, bloom, become dormant, and eventually die. This life cycle moves gradually, over days, weeks, and season to season, but the real barrier to copyright here is not *temporal* but *essential*. The essence of a garden is its vitality, not its fixedness. It may endure from season to season, but its nature is one of dynamic change.").

appear on a screen may not have been generated or even contemplated when the game was written.¹¹¹

In *Microstar v. Formgen*, for example, the court held that a user of a video game infringed by making new map files that called on the game assets to generate new levels of the game.¹¹² The maps were new, so the combination of images the user created didn't exist in the original game. Nonetheless, the court held that the map files infringed because they provided instructions to a computer that created new gameplay using copyrighted images from the original game.¹¹³

111. Although there is player interaction with the machine during the play mode which causes the audiovisual presentation to change in some respects from one game to the next in response to the player's varying participation, there is always a repetitive sequence of a substantial portion of the sights and sounds of the game, and many aspects of the display remain constant from game to game regardless of how the player operates the controls.

Williams Elecs., Inc. v. Artic Int'l, Inc., 685 F.2d 870, 874 (3d Cir. 1982); *accord* *Midway Mfg. Co. v. Artic Int'l, Inc.*, 704 F.2d 1009, 1012 ("The player of a video game does not have control over the sequence of images that appears on the video game screen. The most he can do is choose one of the limited number of sequences the game allows him to choose.").

112. *See* *Micro Star v. Formgen Inc.*, 154 F.3d 1107, 1114 (9th Cir. 1998).

113. *Id.* at 1112. The court held that the maps were a derivative work of the underlying game. This was a bit unusual, because the maps themselves didn't incorporate any features of the game. But they were encoded to generate gameplay that called on and infringed the copyright in the original game.

A number of copyright plaintiffs in generative AI cases have alleged that even if the model isn't a copy of their works, it is a derivative work. Those arguments have largely been rejected by the courts. *See* *Andersen v. Stability AI Ltd.*, 700 F. Supp. 3d 853, 861 (N.D. Cal. 2023) (dismissing plaintiff's claim that the model itself is an infringing derivative work); *Tremblay v. OpenAI, Inc.*, 716 F. Supp. 3d 772, 778 (N.D. Cal. 2024); *Kadrey v. Meta Platforms, Inc.*, 788 F. Supp. 3d 1026, 1057 (N.D. Cal. 2025) ("The plaintiffs allege that the 'LLaMA language models are themselves infringing derivative works' because the 'models cannot function without the expressive information extracted' from the plaintiffs' books. This is nonsensical."); *Justice v. Uncharted Labs, Inc.*, 2026 WL 1430232 (S.D.N.Y. May 21, 2026). That's because the law has been interpreted to say that a work is a derivative work only if it is substantially similar in protectable expression to the original work – the same standard for making a copy. *See* *Litchfield v. Spielberg*, 736 F.2d 1352, 1357 (9th Cir. 1984) (a work is a derivative work only if "it would be considered an infringing work if the material which is has derived from a prior work had been taken without the consent of a copyright proprietor of such prior work," and "to prove infringement, one must show substantial similarity" to that prior work (citation omitted)). That means plaintiffs can't use the derivative works right as an end-run around the limits of the right to make copies.

Some have called the derivative works right "completely superfluous" as a result. 2 Melville Nimmer, Copyright § 8.09[A] (1981). But that's not entirely correct. A derivative work can involve a change to a physical copy of a copyrighted work rather than the creation of a new work. *See, e.g.*, 17 U.S.C. § 101; *Mirage Editions, Inc. v. Albuquerque A.R.T.*, 856 F.2d 1341, 1342–43 (9th Cir. 1988).

Another possible difference – one that is relevant for our purposes – is that the statute nowhere says that a derivative work must be fixed, as a copy must. *See* 17 U.S.C. § 101 (defining derivative work as "a work based upon one or more preexisting works, such as a translation, musical arrangement, dramatization, fictionalization, motion picture version, sound recording, art reproduction, abridgment, condensation, or any other form in which a work may be recast, transformed, or adapted."). Nonetheless, courts have held that there must be some sort of quasi-fixation to create a derivative work. *Micro Star v. Formgen Inc.*, 154 F.3d 1107, XX (9th Cir. 1998) (reaffirming that because "all of the Copyright Act's examples of derivative works took some definite physical form . . . this was a requirement of the act."); *Galoob Toys, Inc. v. Nintendo of Am., Inc.*, 964 F.2d 965, 967 (9th Cir. 1992) ("A derivative work must incorporate a protected work in some concrete or permanent 'form.'"). Were it otherwise, acts like speaking aloud or reading an email on a computer screen could be viewed as the creation of ephemeral

That is a form of non-deterministic output that is somewhat closer to what we see with generative AI. But it is nonetheless different in two important respects.

First, the content produced by playing a video game may not be contained in the same form in the game itself, but the content it will output is generally deterministic. I can make my character look different than others, but I can only do it by combining exact tools and images in the game itself. And I can make that character do many different things in different places in the game, but each of those actions and places are themselves dictated by the parameters of the game. I can't take my character to a place that doesn't exist in the game or have them do something the rules of the game don't permit.

Games like *No Man's Sky* with procedurally generated levels present a harder question for fixation. In procedurally-generated games, the levels are generated anew – involving a degree of randomness – as the player arrives at a new planet. They don't exist until the player arrives, so they aren't stored in the computer, but they are generated using a known algorithm working from a fixed base of content. There isn't litigation on this, but it seems likely a court would say the procedurally generated levels (much less all the possible ones that *might* be generated) didn't preexist in the game; they were created when the user arrived at a new planet and the game engine generated a planet for them. Thus, planets that might at some later point be created in a game aren't fixed in the game itself.

The non-deterministic fixation issue is unlikely to arise in practice in most video game cases because of the second important distinction: the content I generate when I play a video game is the *output* of the interaction between me and the game, displayed or recorded in some medium. It isn't in the game itself. The reason it infringes is that the output, while new and even unpredictable in a sense, copies identifiable elements of the game that *are* present in the computer in a fixed, deterministic form. That was true in *Microstar*, and it would be true of *No Man's Sky*. We don't need to ask whether the new levels are somehow “in” the game in fixed form, because it is the act of generating them as output that is the infringing use.

Contrast a video game with a word processing program. Microsoft Word contains all the elements necessary to build any text output, including this article, all seven *Harry Potter* books, and an infinite number of works that have never been and may never be created. But we wouldn't say that those works are “in” Word merely because they can be assembled from components present in Word. Those individual components (letters, numbers, punctuation, and even higher-order operations like spelling and grammar correction) aren't themselves

derivative works. See Ochoa, *supra* note 9189 at XX (reading the statute to not require fixation “would render illegal merely imagining a modified version of a copyrighted work”); cf. Talkin', *supra* note 20, at 327–28 (noting but not resolving this issue).

copyrightable. It is the act of combining them into a creative output that generates a copyrightable work.

But even if that weren't true, it would be nonsensical to say that Word somehow contains all works ever written or that could be written.¹¹⁴ It may contain the tools for creating those works, but that doesn't mean it has a copy of the works themselves, any more than the digits of pi or any other infinite sequence necessarily contain *Hamlet* (and all other works of authorship now known or later conceived). A key difference here is what information we need as an input in order to extract a particular output. In a deterministic file storage system, the address or perhaps the name of the file is enough to extract the content of that file. At the other extreme, to say that Microsoft Word or the alphabet encodes *Hamlet* or *Harry Potter* isn't meaningful because they are not there in any ordered form from which they reasonably can be extracted. Yes, the works are "in there" in some, very generous sense, but you would need to actually write the works themselves in order to extract them; you'd have to input the information in order to output it.

C. Applying Non-deterministic Analysis to Generative AI

Generative AI sits at an odd space between these examples. LLMs generate text on the basis of probabilistic word and context relationships that are encoded in their model weights.¹¹⁵ Those relationships are influenced by the works the models train on. So an LLM encodes certain word relationships as grammatical and others as nonsensical based on its training. But an LLM isn't simply a neutral collector of word relationships. The more heavily represented particular text strings are in training data, the more likely the LLM will be to replicate those word strings verbatim or nearly verbatim.¹¹⁶ Consider Figure 1: in general, there are many plausible, grammatical English-language continuations of "Mr. and Mrs. Dursley"; after all, these are just two names. The fact that Llama 3.1 70B assigns overwhelmingly high probability to the verbatim continuation from the opening line of *Harry Potter and the Sorcerer's Stone* is not explained by any

114. Cf. *Allocating Rights*, *supra* note 10198, at 1208, 1208 n. 94 ("Granting all rights to the programmer would mean that the programmer would automatically own everything the program was *capable* of generating . . . Because it is possible to put the entire English language into a computer program that can construct prose, it would theoretically be possible for the programmer to claim rights in all writings that were – or could be – produced by the program. Such a claim may seem ridiculous, and yet it would be a logical result of automatically awarding rights in whatever the program did or could produce.") (emphasis in original). While Samuelson's focus was on the establishment of rights, fixation is symmetrical, and the same logic applies to the making of fixed infringing copies.

115. *Sag*, *supra* note 4038 (manuscript at 18) (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6319379) ("The knowledge that resides within an LLM is inherently probabilistic."); See Cooper & Grimmelmann, *supra* note 9, at 196–197 ("the probabilistic element is inescapable").

116. See *Deduplicating*, *supra* note 11, at XX.

feature of the English language, but by the model’s memorization of that specific passage from its training data.

We have shown that for certain works, like *Harry Potter and the Sorcerer’s Stone*, 1984, and parts of *When We Were Eight Years in Power*, it’s straightforward to get some LLMs, notably Meta’s Llama 3.1 70B, to deterministically generate all or a substantial part of the work essentially verbatim with minimal prompting.¹¹⁷ These three works are highly memorized by Llama 3.1 70B; the extraction probabilities across each text are over 95% in all locations,¹¹⁸ which we found makes deterministic extraction straightforward. That suggests that, even if there is not a copy of those works stored in Llama 3.1 70B in the same sense that we would store a Word document in a computer, the model has stored a set of instructions from which it is straightforward and deterministic to generate the work as output. And that result is repeatable; running the same queries produces the same basic result. If that is not exactly like a computer program stored in memory, it’s close enough that a court would probably say that there is a copy (or at least near-perfect blueprints for the making of a copy) in the model itself.¹¹⁹

This is also the case for many other works that aren’t memorized to quite the same extent – for example, those that take multiple tries to extract or for which we can extract only isolated fragments of the work. In many cases, deterministic decoding methods can extract works near-verbatim where the extraction probability is only 0.1%.¹²⁰ The point is, a particular work may be

117. See *supra* I.A decoding schemes; I.B greedy / deterministic extraction. We used a deterministic decoding algorithm called beam search.

118. See *supra* re: probabilistic extraction. This 95% number applies to contiguous sequences of 50 tokens: there for a given prompt of 50 tokens, there is a 95% probability of outputting the verbatim following 50 tokens. With respect to 50-token inputs and outputs, such high extraction probabilities exist throughout these works. Notably, we did not observe the same amount of memorization of this book even in other Meta models, including Llama 1 65B and Llama 2 70B.

119. There are many other ways to extract training data that fit this general description, but they all ultimately involve some form of prompting. Ahmed et al. extracted large portions of near-verbatim text from production LLMs – models that are aligned and embedded inside systems, that are interacted with through an API. In some cases, they had to “jailbreak” the models with adversarial prompts to extract training data (e.g., for Claude 3.7 Sonnet and ChatGPT 4.1); in others, no jailbreaking was necessary (Gemini 2.5 Pro and Grok 3). Extracting Books (Production LLMs), *supra* note 13, at XX. Fine-tuning a model on additional data can also expose memorization of data it was previously trained on, for reasons that aren’t particularly well understood. See generally Nasr et al., *supra* note 12 (fine-tuning GPT 4 makes it possible to extract additional pre-training data). More recently, others have shown that fine-tuning on books can enable extraction of other copyrighted books the model was previously trained on, demonstrating memorization of those books in the model. Xinue Liu et al., *Alignment Whack-a-Mole: Finetuning Activates Verbatim Recall of Copyrighted Books in Large Language Models* XX (Mar. 21, 2026) (unpublished manuscript), <https://arxiv.org/abs/2603.20957>. This type of result changes the model, such that the weights are no longer the same as those that are generally available to the public.

¹²⁰ See Cooper et al. (2026), Extractable Memorization From First Principles, *supra* note XX

more weakly memorized than another in a model, but it may still be possible to retrieve both works reliably and with just one prompt to the model.¹²¹

For other works, the opposite seems to be true. Many models won't output all or any significant part of many works on which we know they trained, even if we try thousands or tens of thousands of times, as with our experiments on *Sandman Slim* (Figure 3). It seems clear that there is not a copy of those works in the model in any meaningful legal sense, at least based on what we can determine using current extraction methods. The model may have been trained on those works, but it apparently didn't retain information about them sufficient to generate them in whole or in part.¹²²

There is also a middle class of works for which it may only be possible to extract significant fractions of the work probabilistically, rather than deterministically.¹²³ We may need to run hundreds or thousands of prompts before the work is extracted.¹²⁴ In this case, we would be seeing non-determinism in action. The model would generate tokens based on its encoded probability of those tokens following existing ones, a probability it has learned from training. But it wouldn't generate the same result every time. So the model is like an enormous decision tree, where each token position is new branching point that can select a different possible token. The model may follow different branches of the tree each time; only the particular branch that happens to correspond to the book will generate text from the book.¹²⁵ The other branches may themselves generate content that is substantially similar to the book, or they may generate new and previously unknown content.¹²⁶ Many of the outputs generated may not be infringing.¹²⁷

121. In the distribution the model assigns, 0.1% may be close to the highest probability sequence for a given prompt; deterministic decoding algorithms like beam search, which approximate identifying the highest probability sequence, will therefore often return such sequences. Near-Verbatim, *supra* note 13, at XX. See Extractable Memorization From First Principles, *supra* note XX.

122. We say "apparently" because, as we noted above, our methods for extracting text and therefore proving memorization in models are imperfect and in development. That is especially true for closed-source models for which scholars need to use adversarial extraction techniques. See generally Extracting Books (Production LLMs), *supra* note 13. Indeed, our own prior work has significantly expanded the universe of provable memorization. See generally Extracting Books (Open-Weight), *supra* note 12; Near-Verbatim, *supra* note 13. It is possible, and even likely, that methods developed later will be able to generate more extraction than we can with today's methods.

123. These would be cases where using a deterministic algorithm like beam search would fail to return the memorized sequence. See *supra* note ~~117~~~~44~~ and surrounding discussion. In this case, we could use a stochastic decoding algorithm, which introduces non-determinism in outputs.

124. These training data are still memorized in a technical sense, but intuitively are not memorized as strongly as the cases where we can extract the work in just a handful of attempts.

125. This picture is also consistent with Figure 1. It's just that one branch (the memorized one) is so high probability that common decoding algorithms mean it is possible to deterministically generate the text.

126. Near-Verbatim, *supra* note 13, at XX.

127. As our work has shown for the verbatim case, this is an open empirical question that depends on the model and work. Extracting Books (Open-Weight), *supra* note 12, at XX. In some cases, most of the outputs may be non-infringing. In others, a branch of (near-)verbatim extraction with 1/1000 probability may be deterministically decodable, depending on how low the probabilities are of the other

In this particular set of circumstances, it may be hard to say there is a “copy” of the work in the model in the sense the Copyright Act means the term without saying there is a “copy” of every branch of the enormous-dimension tree, most of which don’t actually exist in the outside world. From one perspective, if it takes 1000 tries at a prompt to generate verbatim output, the 999 tries that are *not* copies aren’t somehow pre-stored in the AI model.¹²⁸ To say otherwise, we would have to say that a model small enough to run in a single data center (or sometimes on a single computer) somehow stores more unique outputs than there are atoms in the observable universe. That is akin to saying that Microsoft Word stores all the things that could be written with it in a “tree” of potential letters following each other. Yes, it’s possible to generate those things from the model, but it doesn’t mean they are all sitting in the model already made. They are being generated on the fly. But, if that is true of noninfringing outputs generated by the probability tree, it seems wrong to say the result is any different for the one output that happens to infringe. The output, if and when it is generated, may be a copy, but the untapped potential for generating it isn’t a copy.

But that conclusion may depend on the underlying sequence probabilities. One possibility is that the memorized output has an extraction probability of about 1/1000 (0.1%), and the other 999 non-memorized outputs might have substantially lower generation probabilities (e.g., 1/100,000,000, or 0.000001%). If we took another sample of 1000 tries, we would perhaps return the memorized sequence again once and 999 very different non-memorized outputs of similarly low probability. There is clearly something meaningfully different between the single probabilistic extraction case and the other outputs the model generates, even if it takes on average 1000 tries to extract the memorized training data. From this perspective, like the other, it’s also clear that the 999 tries that are *not* copies aren’t somehow pre-stored in the AI model; but it’s less clear that the one try that yields the memorized work isn’t a copy.

If this type of probabilistic case were deemed not a copy, that doesn’t mean the model hasn’t memorized the work. The model has trained on a work, has clearly learned things from that work, and it’s possible that it will produce that

branches. (If 1/1000 is by far the highest probability branch in the tree, common deterministic decoding algorithms like beam search will return it every time.) So even the 1/1000 chance number carries important nuance. See *supra* Part I.A for more on decoding. See Cooper & Grimmelmann, *supra* note 9, at Part I.

128. This assumes the other tries aren’t also similar enough to be infringing; most of the near-verbatim text might be treated the same as the verbatim text for these purposes. So our work showing that there is more near-verbatim memorization than verbatim memorization expands the universe of cases that look more like *Harry Potter*. See Near-Verbatim, *supra* note 13, at XX. For additional nuances for this 1/1000 number, see *supra* note 122. See also Cooper & Grimmelmann, *supra* note 9, at Part II.D.2.

work – much more possible than random chance.¹²⁹ From a machine learning perspective it may make sense to say the work is “memorized.”¹³⁰ But that doesn’t mean the model contains a deterministic copy of that work that is fixed in the model “for a period of more than transitory duration,” as the statute requires.¹³¹ The information is there. It is possible with some significant effort to extract it in the order required to replicate a copyrighted work. But if that extraction is just one possibility among thousands, it happens at the moment of extraction. The information necessary to create that extraction is present in the system. But, for probabilistic cases that may take thousands of attempts to realize the memorized training data in outputs, there is likely no copy as the law means the term.

Generation on the fly in response to prompts likely isn’t fixation for a period of more than transitory duration. The legislative history excluded from this concept embodiments that were “captured momentarily in the ‘memory’ of a computer.”¹³² The Ninth Circuit ignored this part of the statute in *MAI v. Peak*,¹³³ but both the Second and Fourth Circuits have now held that buffering for a few seconds does *not* create a copy that exists for more than a transitory duration even though it is long enough to be perceived.¹³⁴ Permanent storage in computer memory is fixation. And outputting a copy of the work creates a new, fixed copy. But the process of generating results on the fly in the computer doesn’t fix a copy in the model itself.

Our ultimate (and tentative) conclusion is thus mixed. We view a sufficient ease of generation as evidence that the work as a whole is in fact latent in the model and not just being generated through probability. The ability to

129. An old adage holds that if you give enough monkeys enough typewriters they will eventually produce the works of Shakespeare by chance. Similarly, presumably given enough runs a model might generate by happenstance text identical to a copyrighted work even if it hadn’t memorized that work. But in fact, our prior work has shown that the kinds of probabilistic copying we discuss are many, many orders of magnitude more likely than could be explained by random chance. *Extraction*, cite. Extracting Books (Open-Weight); Extractable Memorization From First Principles. Indeed, we have found that the “monkey-at-the-typewriter” events for books correspond to probabilities ranging from 10^{-27} to 10^{-16} (where, as a point of reference, 10^{-3} is 0.1%). As Cooper et al. (Extracting Books Open Weight), note “To give a sense of just how small [the larger of the two (10^{-16}) is, the chance of winning the Powerball lottery jackpot in a single draw is about 1 in 292 million—roughly 34 million times more likely.”

130. Cooper & Grimmelmman, *supra* note 9, at 198. See also our giant appendix E in *Extracting Books (Open-Weight)*, *supra* note 12.

Cooper and Grimmelmman say that “the probability of generating a near-exact copy of a piece of training data has to be high enough to rule out coincidence.” *Id.* at 198. That is right if the goal is to show memorization. And as they note, a one in 1000 chance of generating the work in response to a prompt that tries to elicit it is far, far more than a truly random chance. But the fact that information sufficient to generate part of all of the work is latent in the system is different than saying that a copy persists in the system.

131. 17 U.S.C. § 101.

132. H.R. Rep. No. 94–1476, at 53. Similarly, projecting an image on a movie screen for 1/24th of a second is enough for us to perceive it, but clearly doesn’t count as fixation. *Id.*

133. 991 F.2d 511 (9th Cir. 1993).

134. *Cartoon Network LP v. CSC Holdings, Inc.*, 536 F.3d 121, 128–30 (2d Cir. 2008); *CoStar Grp., Inc. v. LoopNet, Inc.*, 373 F.3d 544, 556 (4th Cir. 2004).

deterministically generate the first *Harry Potter* book using Llama 3.1 70B from a simple seed prompt of ground-truth text from the book suggests that the model has stored instructions that correspond sufficiently to the book that it is akin to the storage of a computer program or video game content. But for books that can be extracted only through a high number of efforts producing different outcomes, what is stored is not a copy of the book itself, but a set of ingredients that can produce copyrighted content, but only when combined in just the right way. A collection of ingredients is not an end product.

That conclusion means that we cannot simply say categorically that models do or do not contain copies of a work in the copyright sense. Some models contain copies of some works under our reading of the statute.¹³⁵ No models (so far as we know) contain copies of all the works on which they are trained. This is possible in degenerate cases that don't reflect state-of-the-art training practices;¹³⁶ but it's impossible for models like Llama 3.1 70B, which are trained on datasets that are far too large for every piece of training data to be losslessly compressed in a model of that size. Ultimately, the question is an evidentiary one that may differ for each model and for each work within that model.¹³⁷ Furthermore, the answer may change, both as new models are developed (ours and others' research suggests, not surprisingly, that larger models memorize more than smaller ones)¹³⁸ and as new ways of assessing the ease of extraction are developed, producing new evidence.

D. The Consequences of Defining a Copy

The consequences of concluding that model contains a copy of a work as the Copyright Act defines that term are potentially significant.

First, if we conclude the model contains a fixed copy and doesn't just generate one when prompted, that copy is *prima facie* an act of infringement whether or not it is ever output or used in any way.¹³⁹ Whatever copy exists in

135. Carlini et al. estimated that a 6 billion parameter model memorizes 1% of its dataset. *Supra* note 8078, at 1. Other estimates range from 0.1% to 10% depending on the model and the technique used. See *Deduplicating Training Data*, *supra* note 7279, at XX. Larger models may memorize more, and previous tools for identifying memorization may undercount it, so that number is likely to increase.

136. See Cooper & Grimmelmann, *supra* note 9, at XX.

137. Cooper & Grimmelmann, *supra* note 9, at 190–91. See also Carlini Blog, *supra* note 15, at XX (“models sometimes emit training data. It doesn’t happen every time. Models can produce output that is not just a direct copy of what they were trained on. And it doesn’t happen never. But sometimes, models directly output text or images that they were trained on.”).

138. See *Extracting Books (Open-Weight)*, *supra* note 12, at XX; Carlini et al., *supra* note 8078, at 2633; Hayes et al., *supra* note 7068, at XX; Near-Verbatim, *supra* note 13, at XX.

139. See *Walker v. University Books*, 602 F.2d 859, 864 (9th Cir. 1979) (“the fact that an allegedly infringing copy of a protected work may itself be only an inchoate representation of some final product to be marketed commercially does not in itself negate the possibility of infringement.”); *Sega of Am. v. Accolade, Inc.*, 975 F.2d 1510 (9th Cir. 1992) (“the Copyright Act does not distinguish between unauthorized copies of a copyrighted work on the basis of what stage of the alleged infringer’s

the model can be “reproduced” even if it is never “perceived” or “communicated.”¹⁴⁰ As a policy matter we might not want to include copies that can never be viewed or used as within the scope of copyright,¹⁴¹ but courts have generally rejected the “communicative function” theory of fixation.¹⁴² So if we decide that anything that might be memorized is a copy for legal purposes, the model itself is an unlawful copy of each of those works unless protected by a defense or exception.

Further, a copy doesn’t have to be of the entire work to infringe. Copying even a small part of the work is sufficient as long as the copy includes a more than de minimis amount of protectable expression.¹⁴³ So while our prior work has shown that for many works it’s possible to extract only some specific portions, that fact doesn’t mean a model wouldn’t infringe that work. If we conclude that that partial extraction possibility makes the model a copy of those parts of the work, the model would still infringe copyright by copying those parts.¹⁴⁴ And even portions of works that can’t be extracted verbatim can often be extracted with very small changes – they can be extracted near-verbatim.¹⁴⁵

If models contain copies of some of the works they train on as the law defines that term, the models themselves *are* infringing copies. That would mean that every new copy of a model also makes a copy of the copyrighted works contained in the model. That has significant consequences for open-weight models. For proprietary systems like those developed by OpenAI and Anthropic, those models exist behind closed doors. There are doubtless numerous copies of the model within the company, and each of those might be infringing, but the company does not widely distribute the models themselves. By contrast, open-

work the unauthorized copies represent”); *MAI Systems Corp. v. Peak Computer, Inc.*, 991 F.2d 511, XX (9th Cir. 1993); *Thoroughbred Software Intern., Inc. v. Dice Corp.*, 488 F.3d 352, 359–360 (6th Cir. 2007) (calculating damages for unused copies of software).

140. 17 U.S.C. § 101.

141. See, e.g., Jessica D. Litman, “Fetishizing Copies.” In *Copyright in An Age of Limitations and Exceptions*, edited by R. Okediji, 107–31. Cambridge: Cambridge Univ. Press, 2017; *UMG Recordings, Inc. v. Shelter Capital Partners LLC*, 718 F.3d 1006, 1019 (9th Cir. 2013) (“[T]he language of the statute recognizes that one is unlikely to infringe a copyright by merely storing material that no one could access . . . [the statute suggests] that if the material were still being stored by the service provider, but was inaccessible, it might well not be infringing.”)

142. Jay Dratler analogizes the “communicative function” argument to Count Dracula, because it “keeps emerging from the coffin, despite repeated interment.” Jay Dratler, *Intellectual Property Law* § 5.03[1][b], at 5–63. See also David G. Luetgen, *Functional Usefulness vs. Communicative Usefulness: Thin Copyright Protection for the Nonliteral Elements of Computer Programs*, 4 Tex. Intell. Prop. L.J. 233 (1996).

143. See *infra* at Part II.D.

144. The exceptions will be cases where the things taken are not themselves copyrightable because they are too small, contain only quotes from other sources or uncopyrightable facts, or are otherwise unprotectable. For example, models that do not otherwise memorize portions of a book often memorize the copyright notice, because it appears repeatedly in standard form across multiple books. But memorizing that basic information would not be copyright infringement.

145. Near-Verbatim, *supra* note 13, at XX. Ippolito et al., *supra* note ~~53~~54, at XX; Extracting Books (Open-Weight), *supra* note 12, at XX 14–15, 184–85; Extracting Books (Production LLMs), *supra* note 13, at XX (near-verbatim in production LLM outputs).

source business models that involve distributing copies of the model weights presumably are making new copies of the works whenever they make a copy of the model. So the potential damages exposure from copyright infringement may be greater for open-source models than for proprietary ones because of the way the models are used. That doesn't make much sense as a policy matter, since there is no reason to think open-source models are more problematic merely because of the way they are distributed, but it seems to follow from a strict application of the law.

Finally, the remedies for copyright infringement aren't limited to damages. If the model is a copy it is presumably subject to remedies like injunction and destruction.¹⁴⁶ There are good equitable reasons not to destroy the models themselves, but plaintiffs have asked for it in numerous cases.¹⁴⁷ And while a less drastic remedy might seem to be removing the copyrighted works from the model, in fact there is nothing simple about "unlearning" in generative AI. As we have previously written, it is very difficult, perhaps even impossible, to remove information from an existing copy of a model, and doing so can have unexpected negative consequences for other aspects of the model.¹⁴⁸ Nor is it evident that we can feasibly identify and come to voluntary licensing deals with those whose works turn out to be memorized in part.¹⁴⁹

All of this assumes that the *prima facie* infringement that would result from declaring a model to contain a copy of a work leads to a finding of liability. But that conclusion doesn't necessarily follow, for two reasons.

First, it is possible that a copy in the model itself will be treated as *de minimis* and therefore noninfringing. To be infringing, the defendant must copy a more than de minimis amount of the work.¹⁵⁰ De minimis copying is normally thought of as copying a small amount of a work. That might apply in some cases

146. 17 U.S.C. § 504, 505.

147. See, e.g., Complaint, *New York Times Co. v. Microsoft Co.*, 777 F. Supp. 3d 283 (S.D.N.Y. 2025) ("Ordering destruction under 17 U.S.C. § 503(b) of all GPT or other LLM models and training sets that incorporate Times Works."); *GEMA v. OpenAI*, 42 O 14139/24 (LG Munchen 11 Nov. 2025)

148. See Machine Unlearning, *supra* note 3028, at at 6–7.

149. See Pamela Samuelson, *Legally Speaking: Does Using In-Copyright Works as Training Data Infringe?*, UC Berkeley Public Law Research Paper, 4 (forthcoming), <https://ssrn.com/abstract=5342516> ("Given the small and uncertain amount each book contributed to the models' value, transaction costs would make licensing infeasible."); Pamela Samuelson, *Justifications for Copyright Limitations and Exceptions*, in *Copyright Law in an Age of Limitations and Exceptions* 12, 38 (Ruth L. Okediji, ed., 2017) (discussing the various ways copyright markets fail to form or are dysfunctional, including transaction costs, disproportionate market power, regulatory interventions, and holdup problems); cf. Xiyin Tang, *The Class Action as Licensing and Reform Device*, 122 Colum. L. Rev. 1627, 1655 (2022) (noting the challenges of finding and licensing AI content).

150. *Castle Rock Entm't, Inc. v. Carol Publ'g Grp., Inc.*, 150 F.3d 132, 138 (2d Cir. 1998) (citing *Ringgold v. Black Entertainment Television, Inc.*, 126 F.3d 70, 75 (2d Cir. 1997)); See generally Oren Bracha, *Not De Minimis: Improper Appropriation in Copyright*, 68 Am. U. L. Rev. 139 (2018) (collecting the law in every circuit).

of memorization, for example where a model memorizes only a short passage from a book.

But *de minimis* has another meaning in the law – not that the copying is of a small amount but that it is “trivial.”¹⁵¹ In deciding whether the copying is trivial, courts consider not just the amount copied but “the observability of the copied work – the length of time the copied work is observable in the allegedly infringing work,” and factors such as “focus, lighting, camera angles, and prominence.”¹⁵² Indeed, courts have noted that “observability” is fundamental to the analysis.¹⁵³ When a work appears momentarily in the background of a film and is not the focus of the scene, for instance, so that the ordinary observer wouldn’t notice it, courts may hold it *de minimis* even though the entire work is depicted.¹⁵⁴

Memorization in a model, even if it is determined to be a copy, may be *de minimis* if it is difficult or impossible to practically extract that copy from the model. Whether that is true is an empirical question that will differ from case to case. It might be difficult to extract a work as a practical matter because the probability of generating a work from a model is so low. If it takes 1,000 tries to extract a work, users probably won’t persist just to extract a work they already have access to the content of. Further, even if the work is heavily memorized in the base model, that doesn’t mean it would be easy to extract from a production system like ChatGPT that has some (however limited) guardrails built in to try to prevent surfacing copies in outputs to users. If the model actually deployed to consumers implements sufficient guardrails, it may be very difficult to extract the memorized work. And if you can’t practically get the work out of the model, courts might view the inaccessible copy as *de minimis*. Courts haven’t confronted such a case as a *de minimis* use, so it would be an extension of the doctrine, but it seems a logical one.¹⁵⁵

151. *Sandoval v. New Line Cinema Corp.*, 147 F.3d 215, 217 (2d Cir. 1998) (quoting *Ringgold v. Black Ent. Television*, 126 F.3d 70, 74 (2d Cir. 1997)); *Gordon v. Nextel Commc’ns & Mullen Advert., Inc.*, 345 F.3d 922, 924 (6th Cir. 2003).

152. *Ringgold*, 126 F.3d at 75 (citing 4 Melville B. Nimmer & David Nimmer, Nimmer on Copyright §§ 13.03[A][2] (1997)); *see also* *Gordon v. Nextel Commc’ns & Mullen Advert., Inc.*, 345 F.3d 922, 924 (6th Cir. 2003).

153. *Sandoval*, 147 F.3d at 217; *Solid Oak Sketches, LLC v. 2K Games, Inc.*, 449 F. Supp. 3d 333, 344 (S.D.N.Y. 2020). One of us has shown that proprietary models vary widely in whether and how they build safeguards against extracting copyrighted works. *Extracting Books* (Production LLMs), *supra* note 13, at XX. For more on the systems vs. models distinctions for extracting memorization, see Cooper and Grimmelmann, *supra* note 9, at Part II.G.

154. *Ringgold*, 126 F.3d at 77. *Cf. Matthews Conveyor Co. v. Palmer-Bee Co.*, 135 F.2d 73, 85 (6th Cir. 1943) (asking “whether so much has been taken as would sensibly diminish the value of the original”); *Gottlieb Dev. LLC v. Paramount Pictures Corp.*, 590 F. Supp. 2d 625 (S.D.N.Y. 2008) (depicting a pinball machine in the background of a movie was *de minimis*).

155. *Cf. Jessica Silbey & Samantha Zyontz, De Minimis Copying: An Empirical Study*, 15 J. Intell. Prop. & Ent. L. 118 (2025) (studying *de minimis* cases and finding that the argument usually fails).

Second, it is possible that the models themselves, like the training datasets from which they were built, will be held a fair use of the works they memorize. U.S. courts have so far held that training models is fair use.¹⁵⁶ Even if a model contains a copy of some of the works it has trained on, that copy too might be fair use, especially if it is latent unless and until someone uses it to generate an infringing output. Many of the same arguments that justify fair use of the training data apply to memorization in the model itself.¹⁵⁷ If the work exists in the model but is never output by the model, so its presence there only affects the weights given to language the model uses when creating new and noninfringing works, the use of those copies is likely transformative for the same reasons it is in model training. And the market effects seem similar.¹⁵⁸ One difference is that the model itself is generally a commercial product (or the basis of a commercial service selling the use of the model), while the training dataset is not, and commerciality has taken on greater significance after *Andy Warhol Foundation v. Goldsmith*, which revived it as an important consideration, rejecting thirty years of contrary precedent.¹⁵⁹

156. *Kadrey v. Meta Platforms, Inc.*, 788 F. Supp. 3d 1026, 1060 (N.D. Cal. 2025), *Bartz v. Anthropic*, 787 F. Supp. 3d 1007, 1034 (N.D. Cal. 2025). See generally Mark A. Lemley & Bryan Casey, *Fair Learning*, 99 Texas L.R. 743 (2021); Sag, *supra* note 4038. The Copyright Office took a different position, but it did so only by articulating a new theory of market harm that is directly at odds with existing caselaw. See United States Copyright Office, *supra* note 17, at 65–66. Compare Edward Lee, *Copyright Dilution Under Constitutional Scrutiny*, 25 Chi.-Kent J. Intell. Prop. 1 (2026) (explaining the flaws with the Copyright Office’s argument). But only district courts have ruled on the issue, and the result could change. For a comprehensive analysis of training as fair use, see Edward Lee, *Fair Use and the Origin of AI Training*, 63 Hous. L. Rev. 105 (2025).

157. See Tian “Tony” Chen, *Language Models’ Verbatim Copying: Copyright Infringement Analysis Through the Lens of The New York Times Co. v. Microsoft*, OpenAI, Inc. et al, 43 Cardozo Arts & Ent. L.J. 349, 364–66 (2025) (arguing that memorization in models should be fair use because it is non-expressive and serves the purpose of producing noninfringing outputs).

158. Some copyright owners and their advocates have suggested abandoning the existing legal definition of market effect and creating a theory of “market dilution,” under which competing with a copyright owner would weigh the fourth factor against fair use, even if – as is normally the case – the competing work is not infringing. Jacqueline C. Charlesworth, *Generative AI’s Illusory Case for Fair Use*, 26 *Vanderbilt Journal of Entertainment and Technology Law* 323 (2025); see United States Copyright Office, *supra* note 17, at 65. That is not the law. Indeed, existing cases flatly reject it. See, e.g., *Sony Computer Ent., Inc. v. Connectix Corp.*, 203 F.3d 596, 607 (9th Cir. 2000) (“[B]ecause the Virtual Game Station is transformative, and does not merely supplant the PlayStation console, the Virtual Game Station is a legitimate competitor in the market . . . For this reason, some economic loss by Sony as a result of this competition does not compel a finding of no fair use.”); Edward Lee, *Copyright Dilution Under Constitutional Scrutiny*, 25 Chi.-Kent J. Intell. Prop. 1 (2026).

159. *Andy Warhol Found. For the Visual Arts, Inc. v. Goldsmith*, 598 U.S. 508, 509 (2023). Compare *Campbell v. Acuff-Rose Music, Inc.*, 510 U.S. 569, 570 (rejecting the presumption that commercial uses were unfair). As Pam Samuelson notes, even after *Warhol* there are many ways commercial uses can be held fair. Pamela Samuelson, *Justifications for Fair Uses*, 2025 Wis. L. Rev. 1047.

Fair use by models has been much discussed elsewhere,¹⁶⁰ and it is not our intention to revisit that debate here. We raise it here only to note that the issue of whether a model contains a copy of a work is not necessarily determinative. Where a model is deemed a copy, resolution of the fair use question takes on even more importance because of the legal consequences we discuss in this section.

It is also possible that some (though not all) works that are now memorized might be licensed. But it is not feasible to simply create a license to all training content, for reasons Pam Samuelson has explained.¹⁶¹ It also doesn't seem possible to eliminate all memorization¹⁶² and similar to "unlearning", even if it were possible, doing so might have negative unintended consequences for model behavior.¹⁶³ And it is not computationally feasible to determine which works each model memorizes in sufficiently relevant part in order to try to narrow down the universe of licenses.¹⁶⁴

E. Policy Implications

The Copyright Act was written in 1976. Its definition of "copy" was intended to encompass technologies that were just emerging at the time, including computers. But Congress did not have in mind, and did not address, the sorts of non-deterministic behavior we see with today's generative AI models. In an ideal world we might rewrite the law to limit copyright to covering copies that actually reach or communicate to people, deleting the word "reproduced" from the "perceived, reproduced, or otherwise communicated" language in the statute. That would exclude the models themselves as copies, but would still encompass model output that is substantially similar to a copyrighted work.¹⁶⁵ But we don't live in an ideal world, and given the composition of

160. See generally Henderson et al., *Foundation Models and Fair Use*, 24 J. Mach. Learning Res. 1 (2023), <https://arxiv.org/abs/2303.15715>; Cooper & Grimmelmman, *supra* note 9.

161. See Samuelson, *Legally Speaking*, *supra* note ____.

162. See e.g. Cooper & Grimmelmman, *supra* note 9, at 204 and surrounding footnotes; Vitaly Feldman, *Does Learning Require Memorization? A Short Tale About a Long Tail*, PROC. 52ND ANN. ACM SIGACT SYMP. ON THEORY COMPUT. 954 (2020). See also Cooper & Grimmelmman, *supra* note 9, at Part II.F.; Talkin', *supra* note 20, at XX; Sag, *supra* note 4038, at XX.

163. See Machine Unlearning, *supra* note 3028, at 6–7.

164. It is feasible to test a potentially very large set of works, as we have done in some of our work. But training datasets are enormous, and it currently isn't feasible to test every model on every work in its respective training set. For instance, Cooper et al. spent over a year of GPU compute testing approximately 0.1% of the books in the Books3 corpus on Llama models, which are known to have been trained on Books3 – and Books3 is only a small fraction of all of Llama's training data. See generally *Extracting Books (Open-Weight)*, *supra* note 12.

165. Haviv et al. suggest that this is already the case, and that the law distinguishes memorization from copyright. *Supra* note 8785, at 4–5. We are sympathetic to the policy argument, but skeptical that it reflects existing law, at least in the United States. Oren Bracha makes a similar argument. Oren Bracha, *The Work of Copyright in the Age of Machine Production*, 38 Harv. J. L. & Tech. 171 (2024). He couches it as consistent with current law, but again, we are less convinced that is true. Courts have rejected the idea that a work must serve a communicative function to be copyrighted, applying the law to unpublished letters, for instance. *Salinger v. Random House, Inc.*, 811 F.2d 90, 94 (2d Cir. 1987); *cf.*

Congress and the power of industry lobbyists, there are serious risks that revisiting the basic structure of the Copyright Act would end up making things worse, not better.

Given the indeterminacy of the Act's definition of "copy" and "fixed," perhaps courts could read the terms in ways that exclude the models themselves as copies but would still encompass model output that is substantially similar to a copyrighted work. The closest analogy is to the cases that have treated temporary buffer copies internet sites make in the course of streaming or preparing to deliver content as not fixed and therefore not copies under the statute.¹⁶⁶ Those cases all involve temporary, intermediate instantiations of a work made in the service of delivery of that work. The instantiations in those cases mostly exist only for a short period of time, unlike the weights in AI models. But the courts seem to be applying the principle that a brief instantiation of a work in the service of generating an end product that can't itself be perceived or used shouldn't be viewed as a fixed copy, and this seems analogous to generative AI. The parallel argument would be that even if research techniques reveal that a model contains a copy, if that copy is never perceived or communicated to the outside world, the law shouldn't condemn it. Alternatively, courts might treat such unobservable works as *de minimis* and therefore noninfringing.

But either approach would require an extension of current law, either by statute or by judicial interpretation. As the law stands now, the answer to the question whether a model that memorizes content from a work is itself a copy of that work is "it depends on the model and the work." That isn't a very satisfying answer, but it is the one the law and the science lead us to.

CONCLUSION

Memorization of copyrighted works in AI models forces copyright law to confront an issue it has so far managed to avoid – what to do about "probabilistic copies" of training works that the model might or might not generate in outputs.

David G. Luetgen, *Functional Usefulness vs. Communicative Usefulness: Thin Copyright Protection for the Nonliteral Elements of Computer Programs*, 4 TEX. INTELL. PROP. L.J. 233, 250–52 (1996) (arguing for narrower protection for noncommunicative works but not contending that there is no protection). Under the statute the same should also be true of infringing copies, since the definition of the terms is identical for both sections.

Nor is it the case, as Haviv et al. suggest, that memorization is generally of "generic, functional, or weakly protected material." Haviv et al., *supra* note 8785, at 8. While that is surely sometimes true, we and others have shown numerous instances of verbatim memorization of creative works. See *supra* I.C footnote for additional discussion.

166. *Cartoon Network LP, LLLP v. CSC Holdings, Inc.*, 536 F.3d 121, 130 (2d Cir. 2008); *UMG Recordings, Inc. v. Shelter Capital Partners LLC*, 718 F.3d 1006, 1016 (9th Cir. 2013); *CoStar Grp., Inc. v. LoopNet, Inc.*, 373 F.3d 544, 556 (4th Cir. 2004). *Cf. Spotify, C-496/24 (CJEU 2026)* (holding that streaming does not create a copy even though it requires temporary instantiations of the files).

We think the most likely answer is a functional one: the model contains a copy of a work if, but only if, the work can be extracted from the model with relatively little effort. That means that whether a model contains a copy of a work will vary depending on the model, the work, and the state of the art of extraction research.¹⁶⁷ We would be better served by a rule that focused not on the metaphysical presence of copies in a model that may never be perceived in the outside world, but on whether the model generates infringing output.

¹⁶⁷ Indeed, over time, extraction science has developed new techniques that, in the instances in which they've been tested, make extraction easier – expanding our knowledge of how much state-of-the-art models memorize.